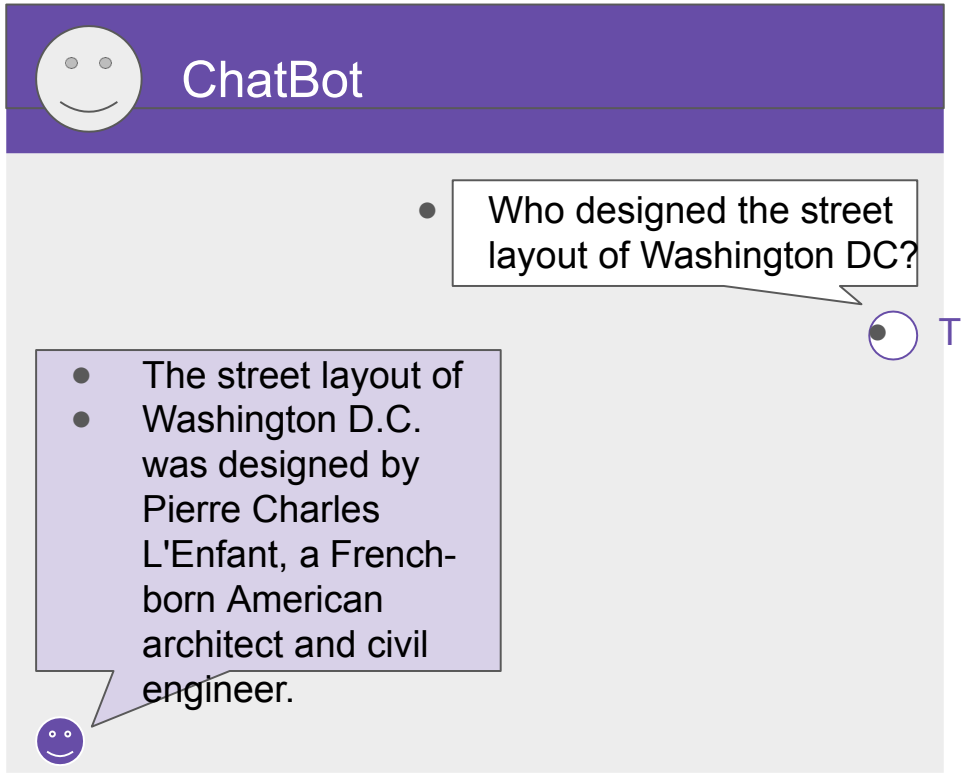


Data Science and Machine Learning

Generative AI

Generative AI & Large Language Models

Generative AI



Generative AI

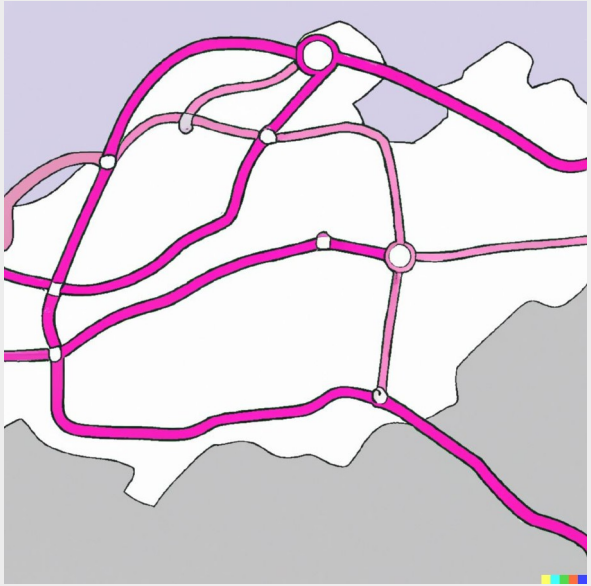
pAIntBox

What do you want to

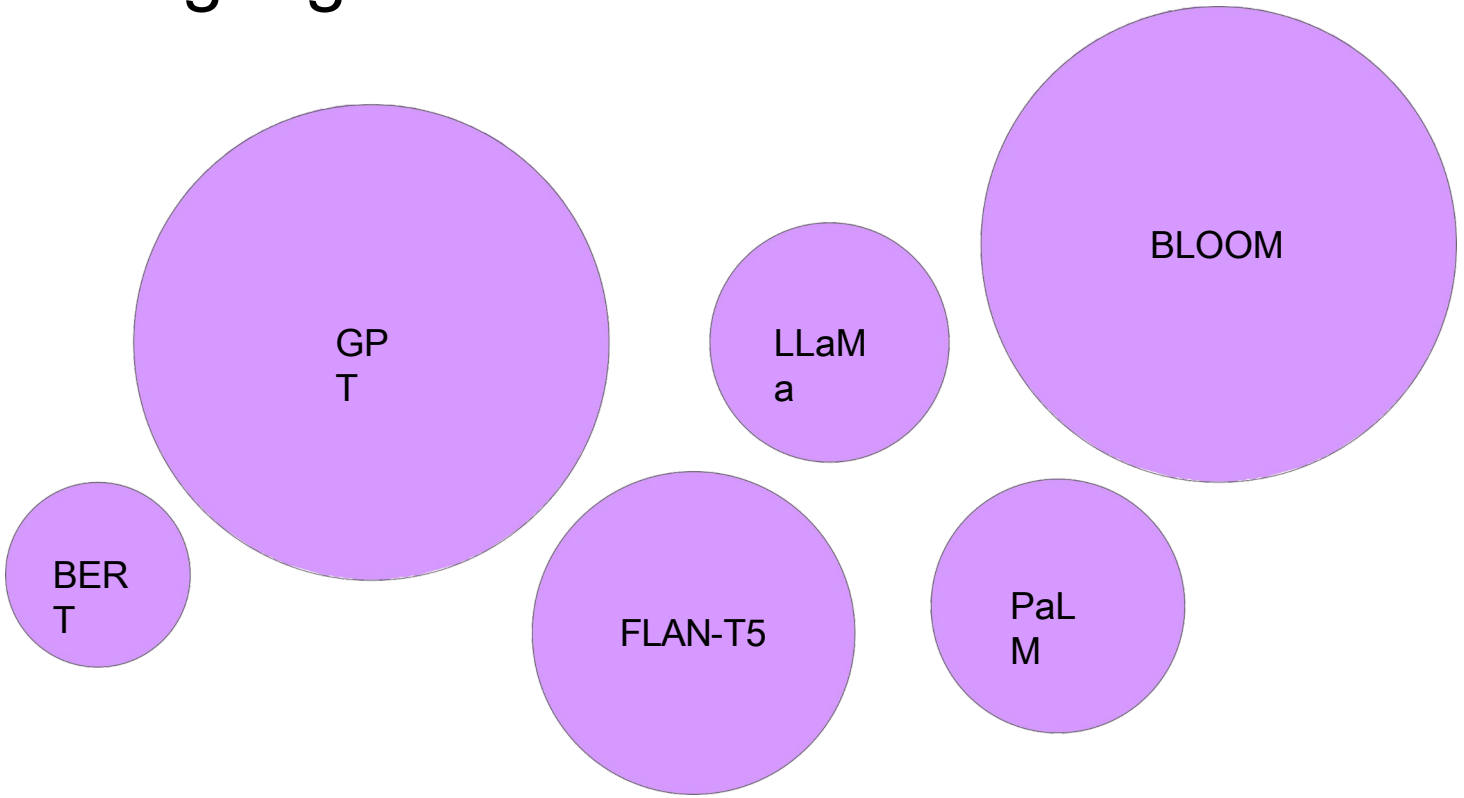
An imaginary subway map in a coastal city.

Image dimensions: by (Max 2048)

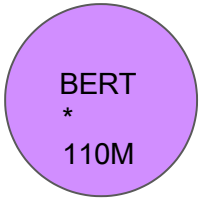
Generate



Large Language Models



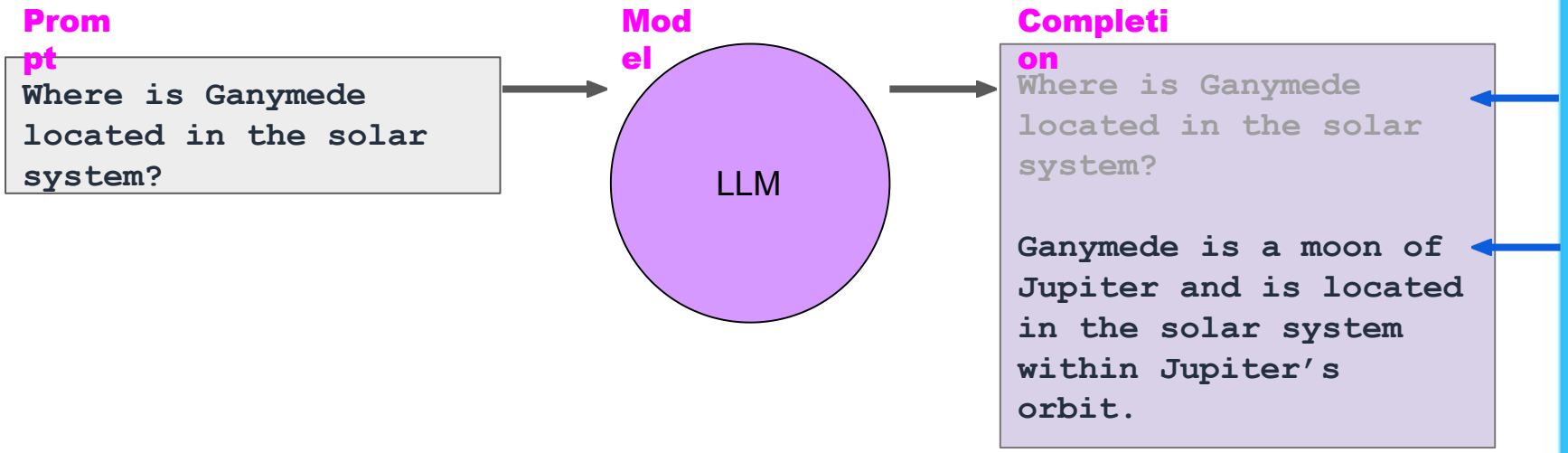
The significance of scale: language understanding



*Bert-base

BLOOM
176B

Prompts and completions

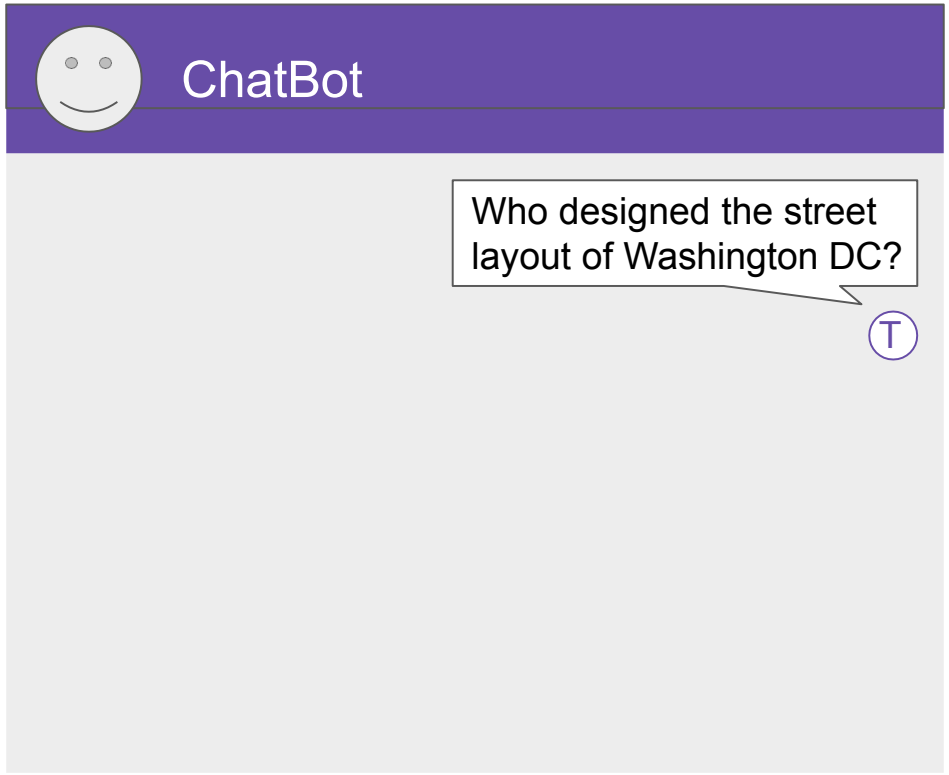


Context window

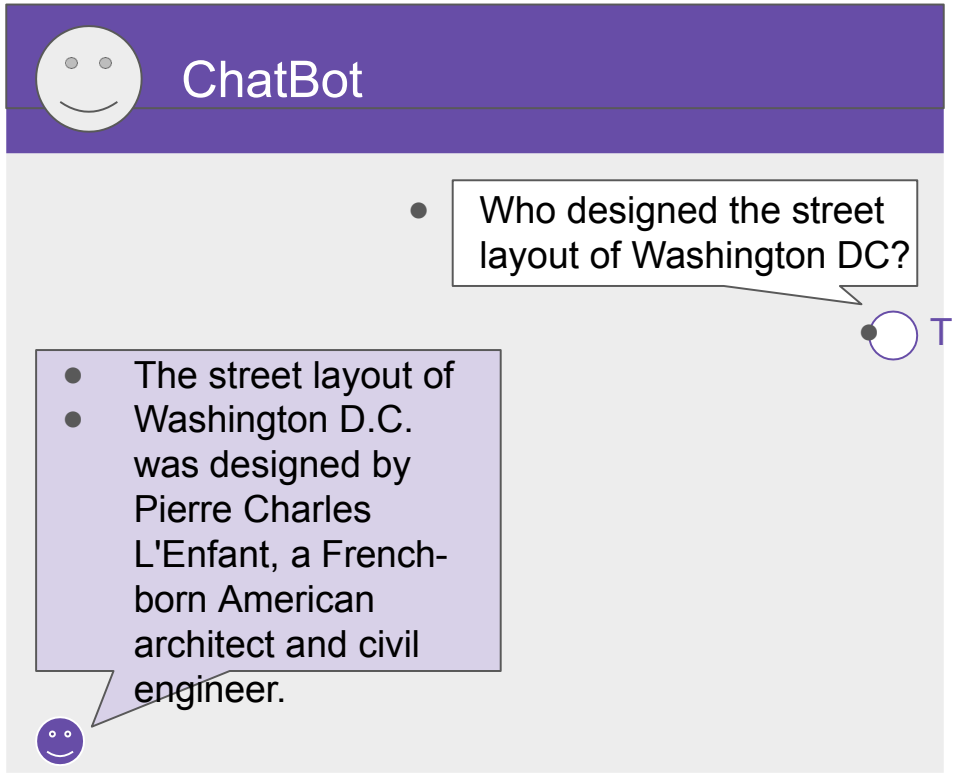
- For exemple, a few 1000 words.

Use cases & tasks

LLM chatbot

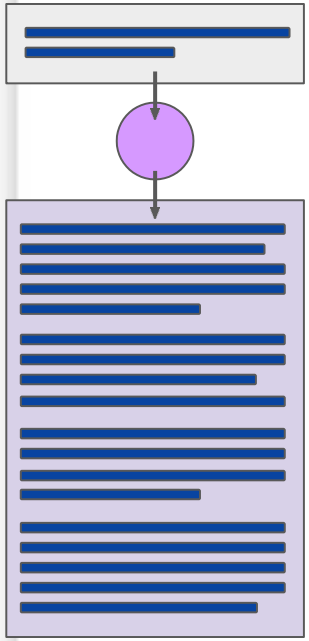


LLM chatbot

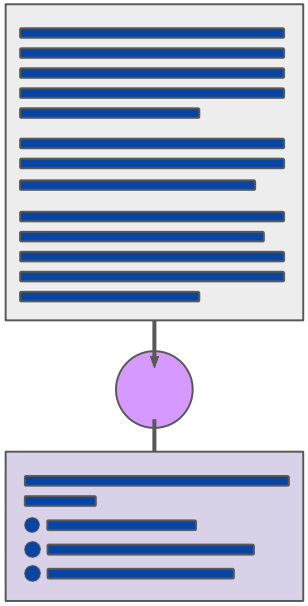

ChatBot
Who designed the street layout of Washington DC?
T
The street layout of Washington D.C. was designed by Pierre Charles L'Enfant, a French-born American architect and civil engineer.

LLM use cases & tasks

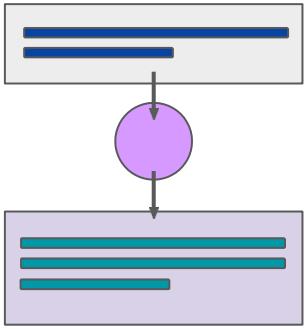
Essay Writing



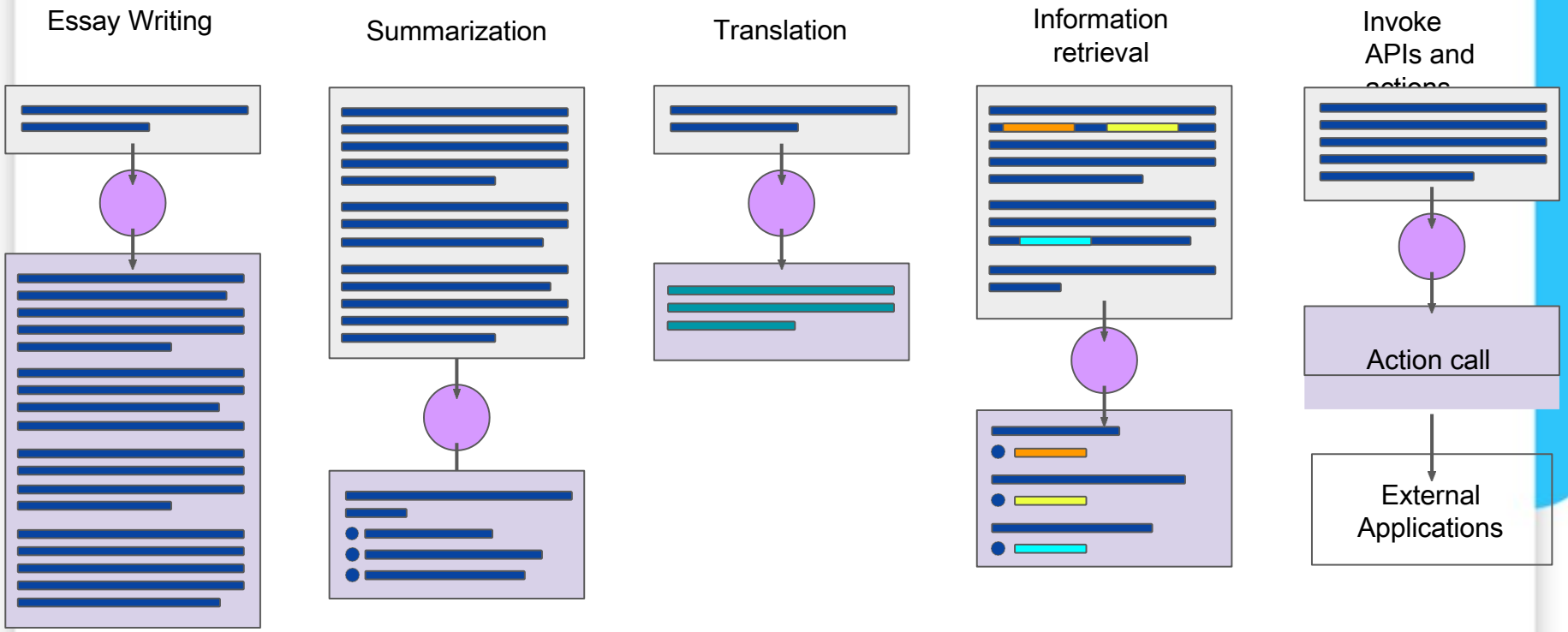
Summarization



Translation

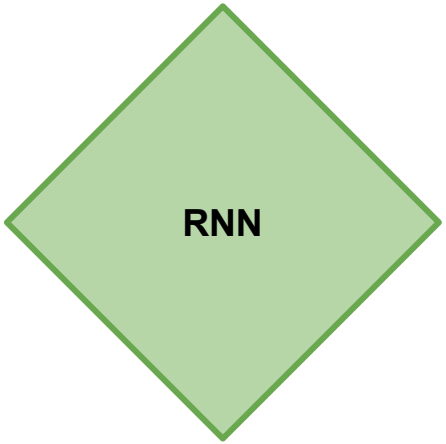


LLM use cases & tasks



How LLMs work - Transformers architecture

Generating text with RNNs



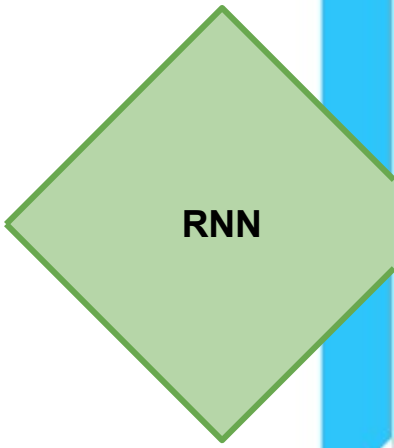
Generating text with RNNs



Generating text with RNNs



my … tea tastes .

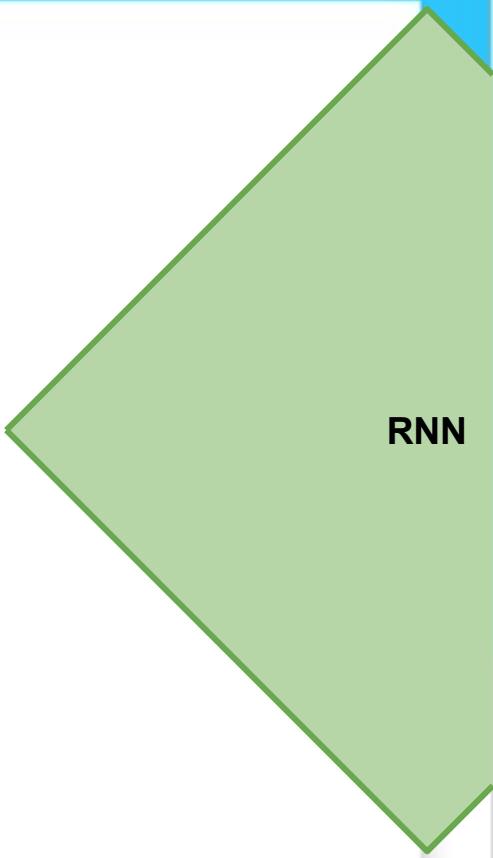


Generating text with RNNs



, my tea tastes

bad … .



Generating text with RNNs



Generating text with RNNs



Understanding language can be challenging

I took my money to the____ bank.



River bank?

Understanding language can be challenging

The teacher's book?

The teacher taught the student with the book.

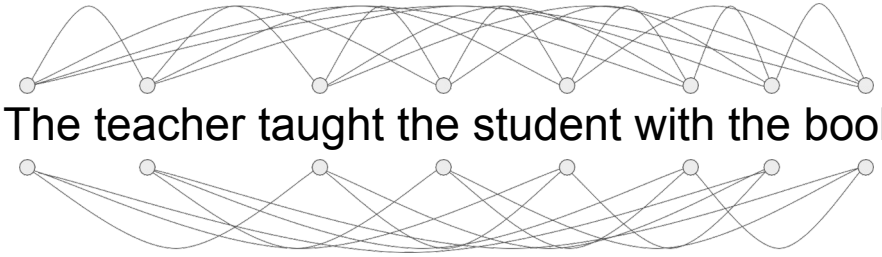
The student's book?

RNN

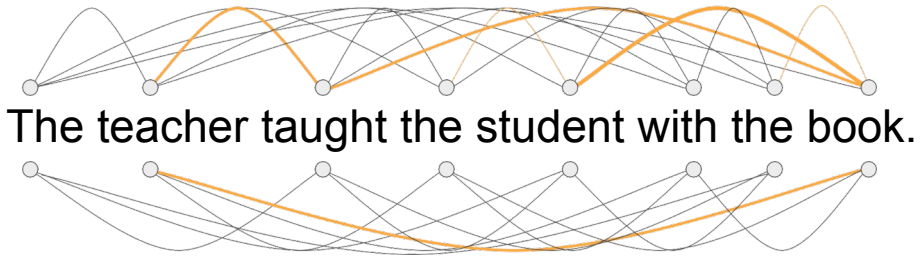


The teacher taught the student with the book.

Transformers



Transformers



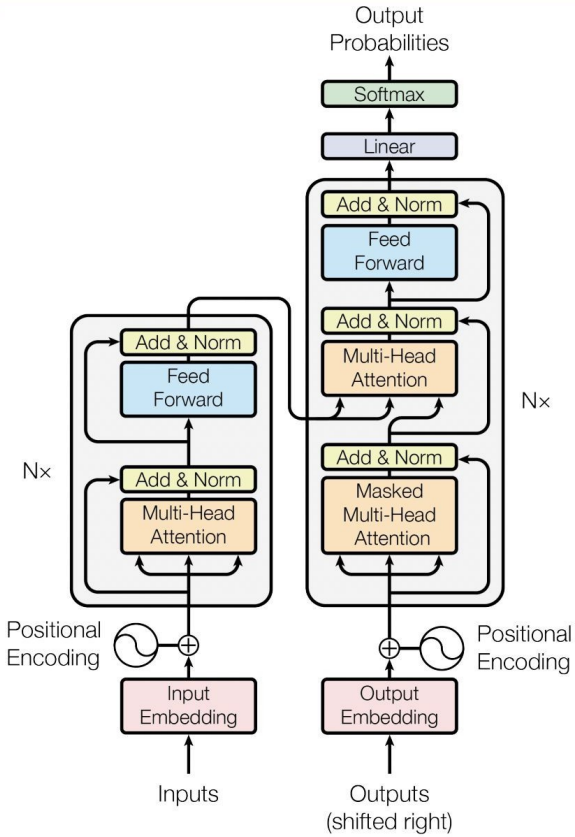
Transformers

Attention Is All You Need

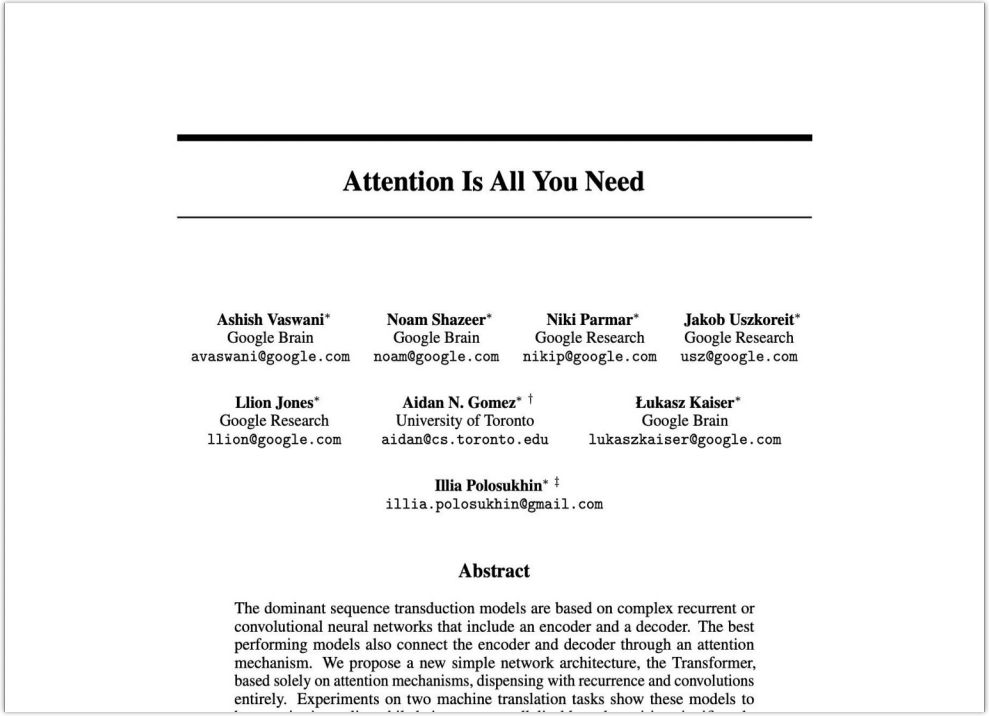
Ashish Vaswani* Google Brain avaswani@google.com
Noam Shazeer* Google Brain noam@google.com
Niki Parmar* Google Research nikip@google.com
Jakob Uszkoreit* Google Research usz@google.com
Llion Jones* Google Research llion@google.com
Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu
Łukasz Kaiser* Google Brain lukaszkaiser@google.com
Illia Polosukhin* ‡ illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to



Transformers



Attention Is All You Need

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Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

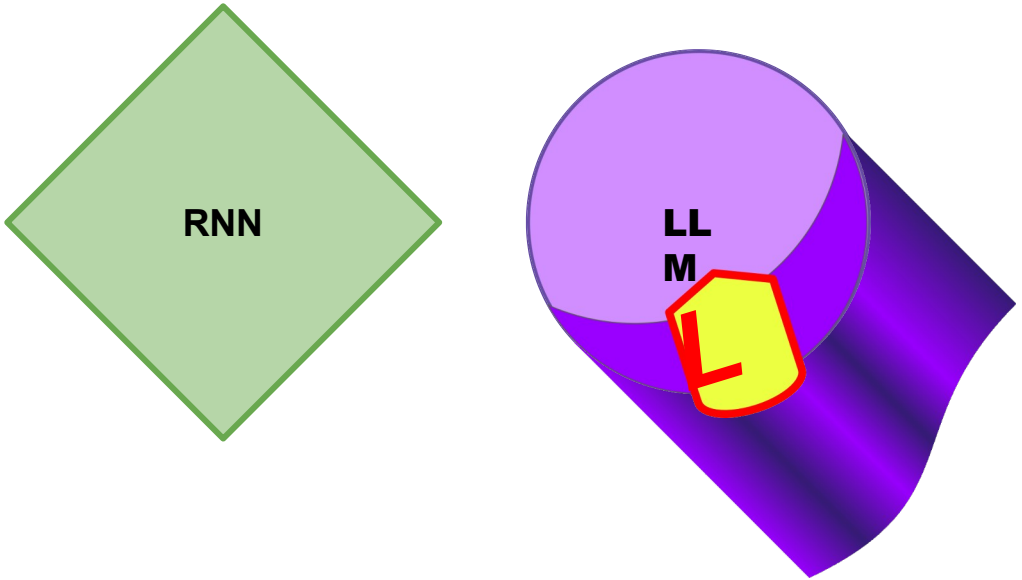
Illia Polosukhin* ‡ illia.polosukhin@gmail.com

Abstract

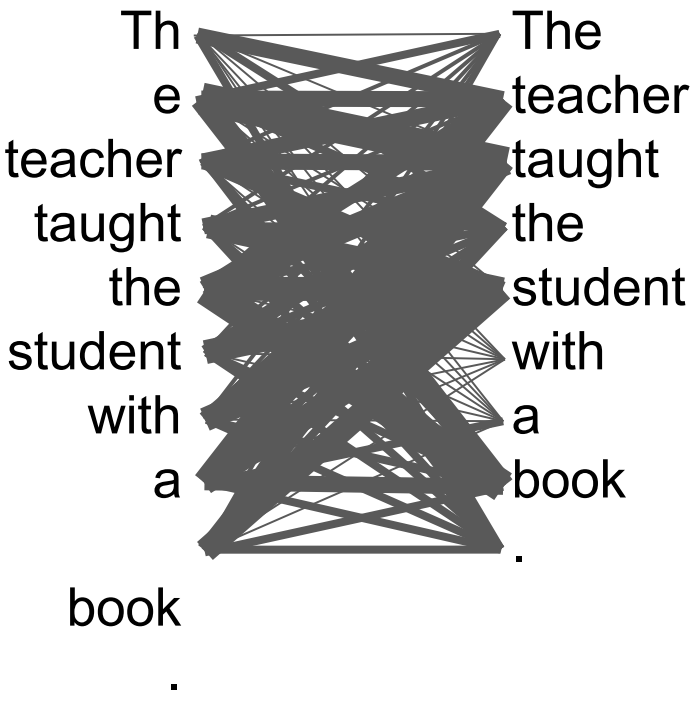
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to

- Scale efficiently
- Parallel process
- Attention to input meaning

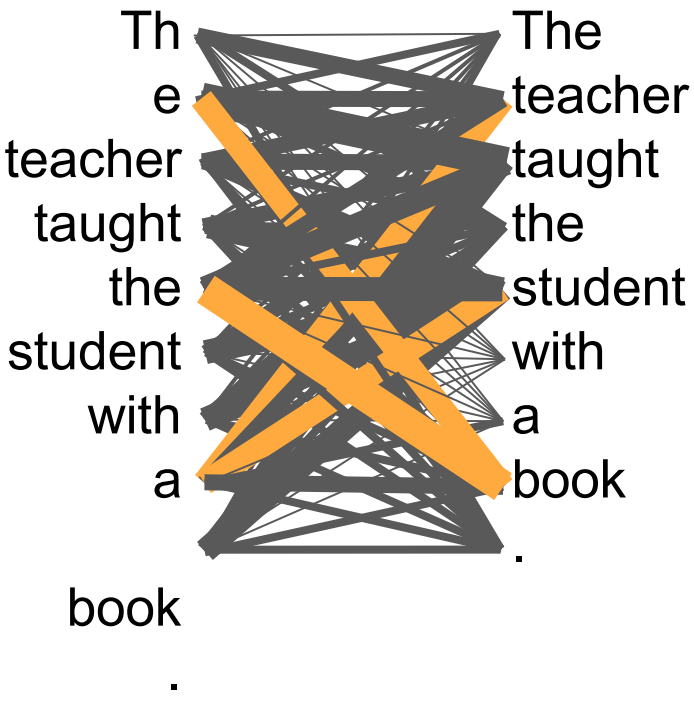
Transformers



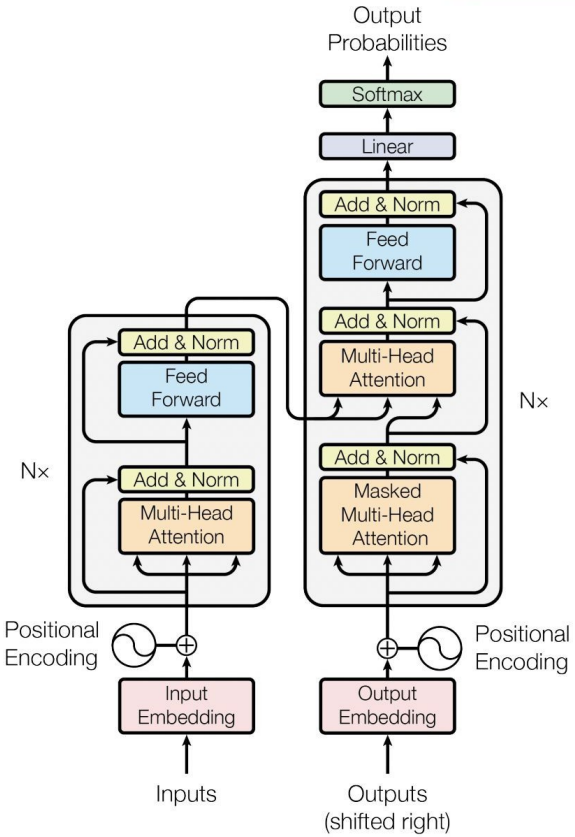
Self-attention



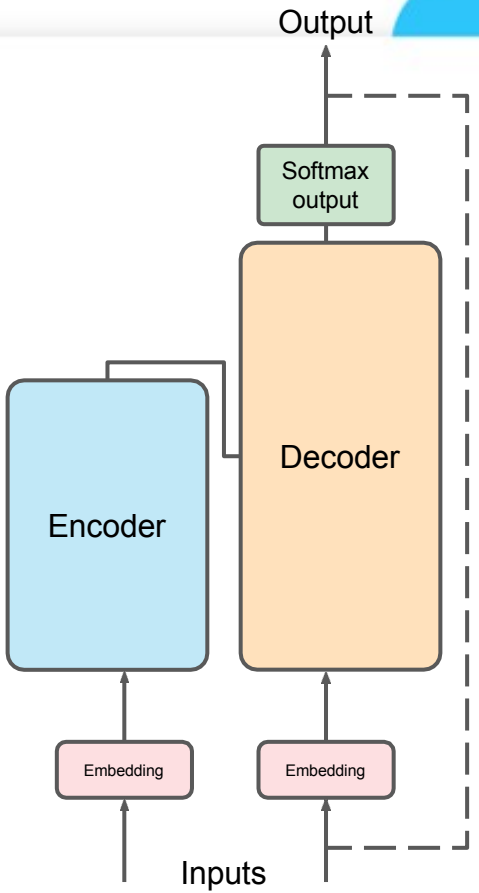
Self-attention



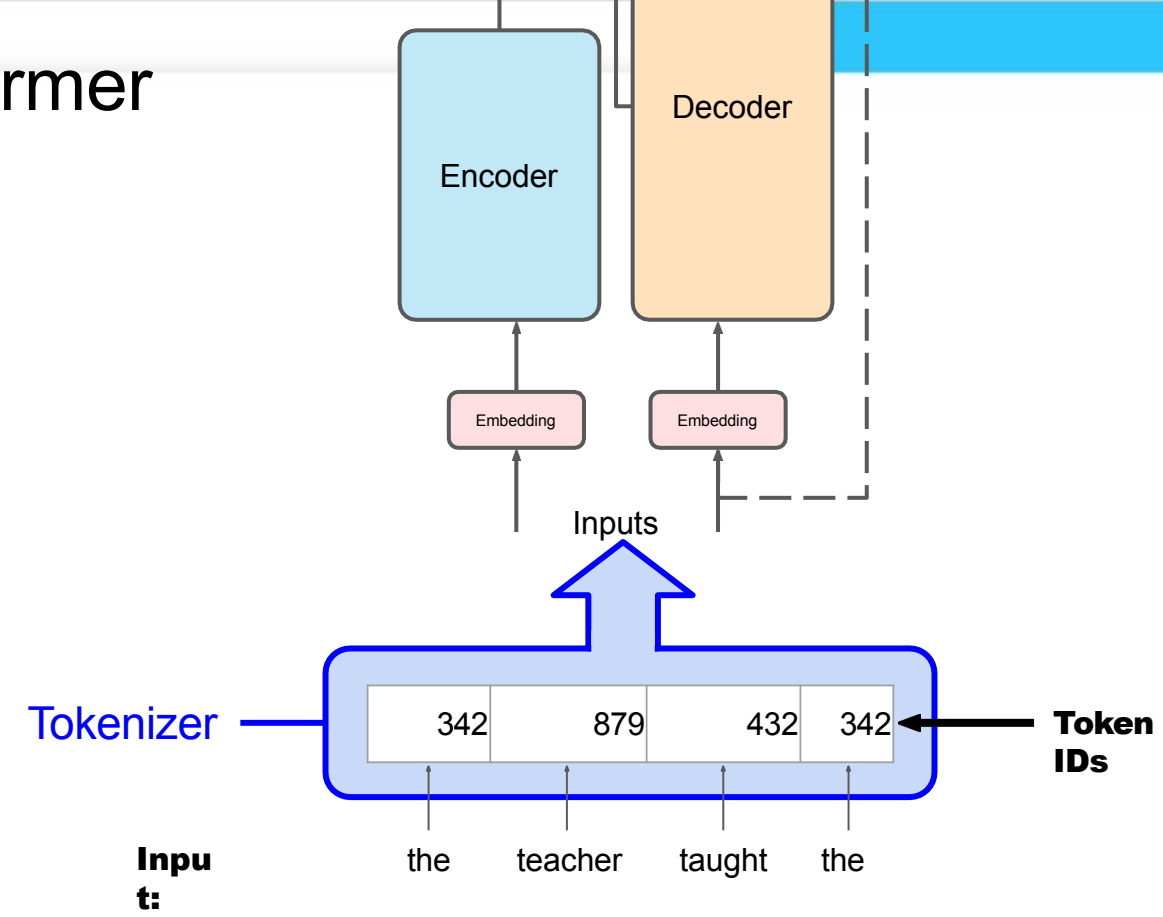
Transformers



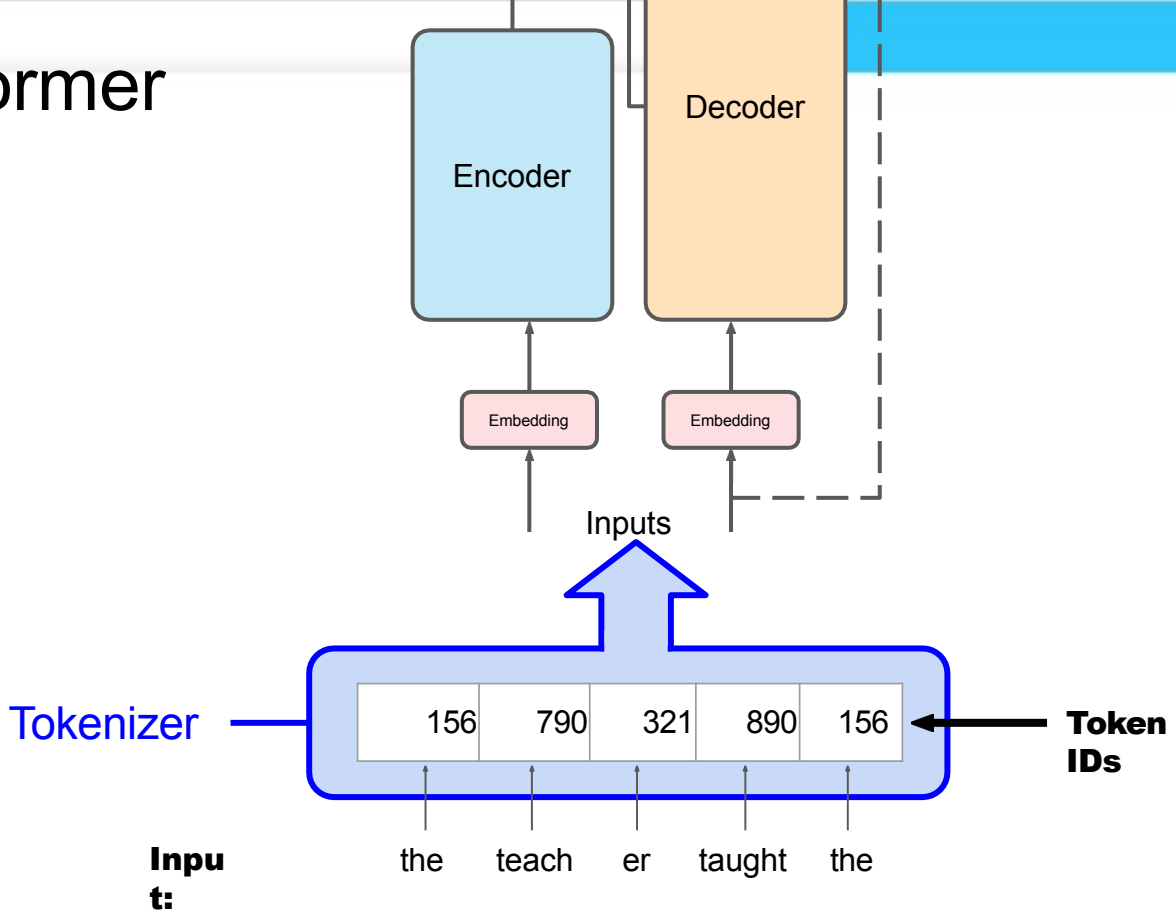
Transformers



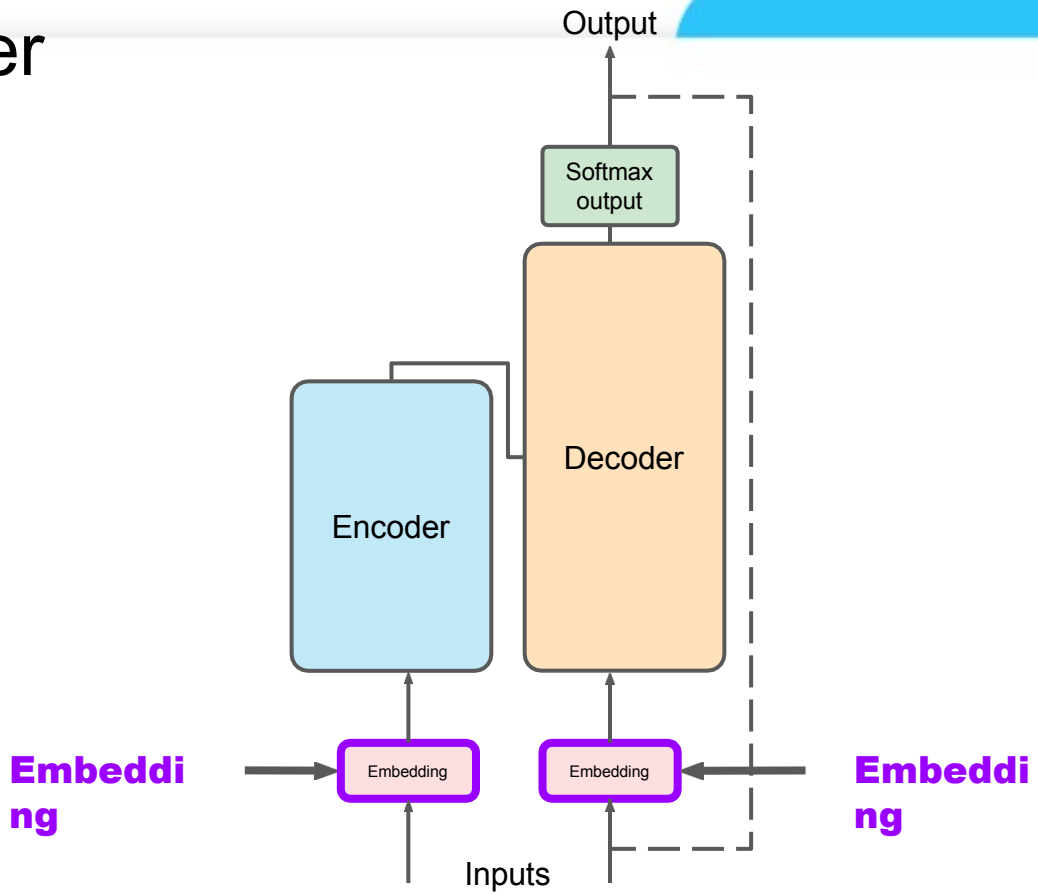
Transformers



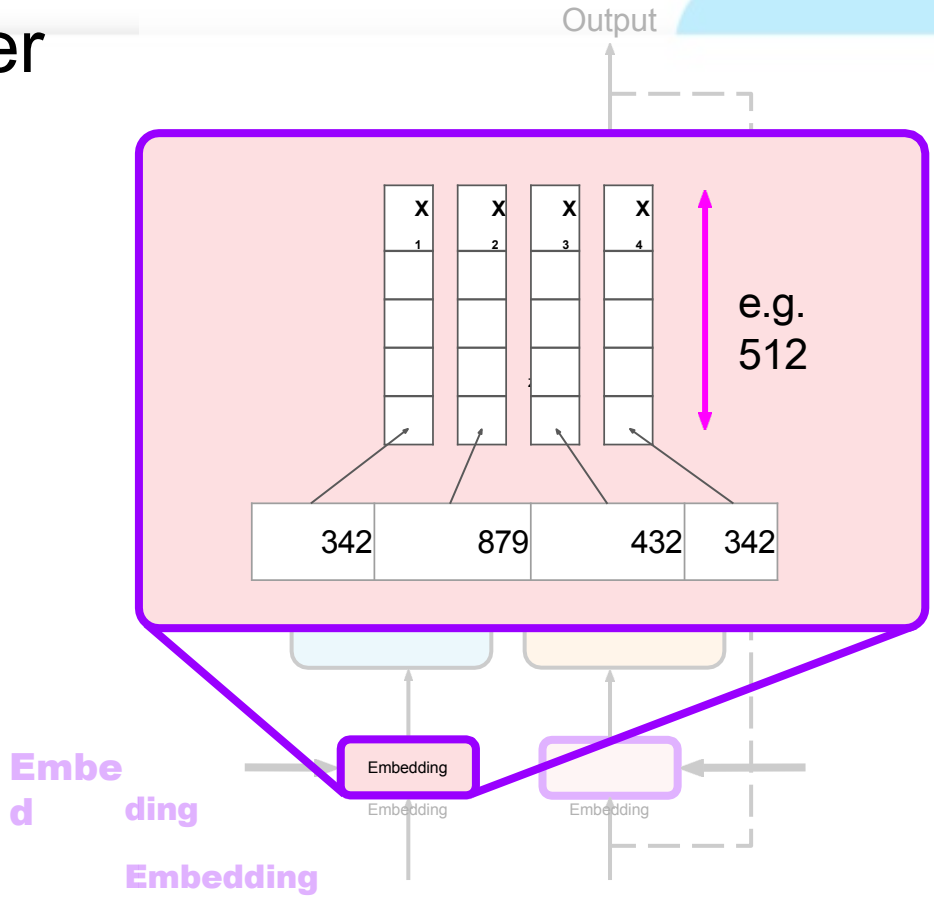
Transformers



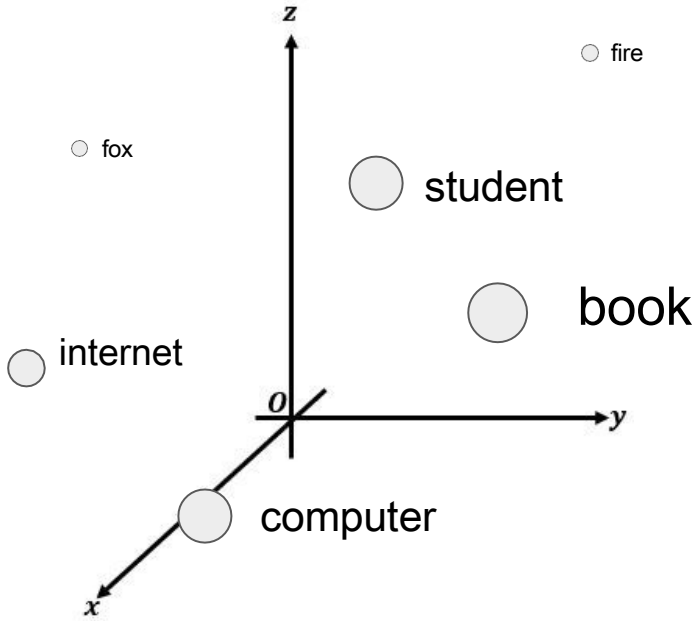
Transformers



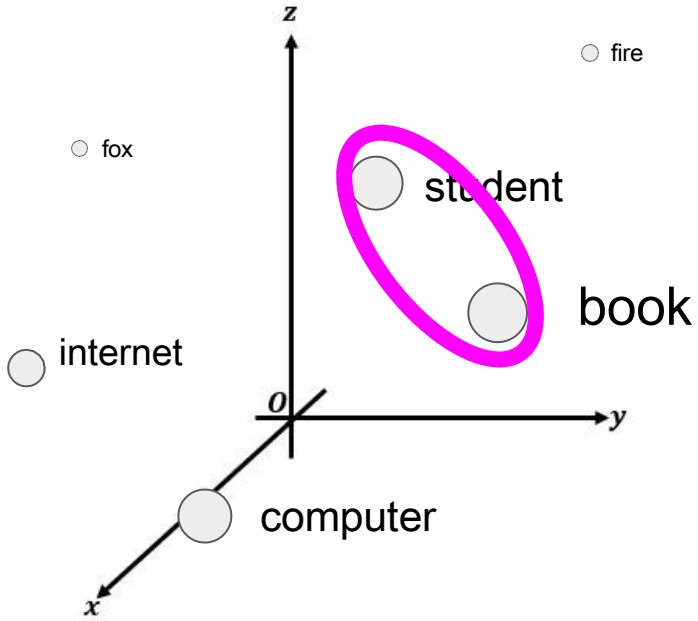
Transformers



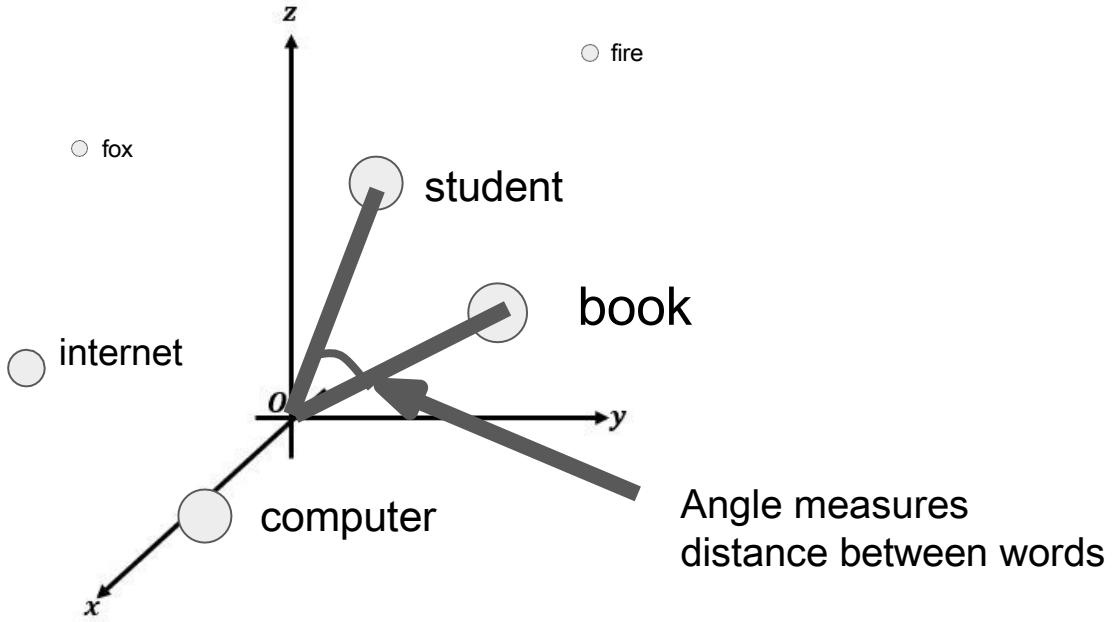
Transformers



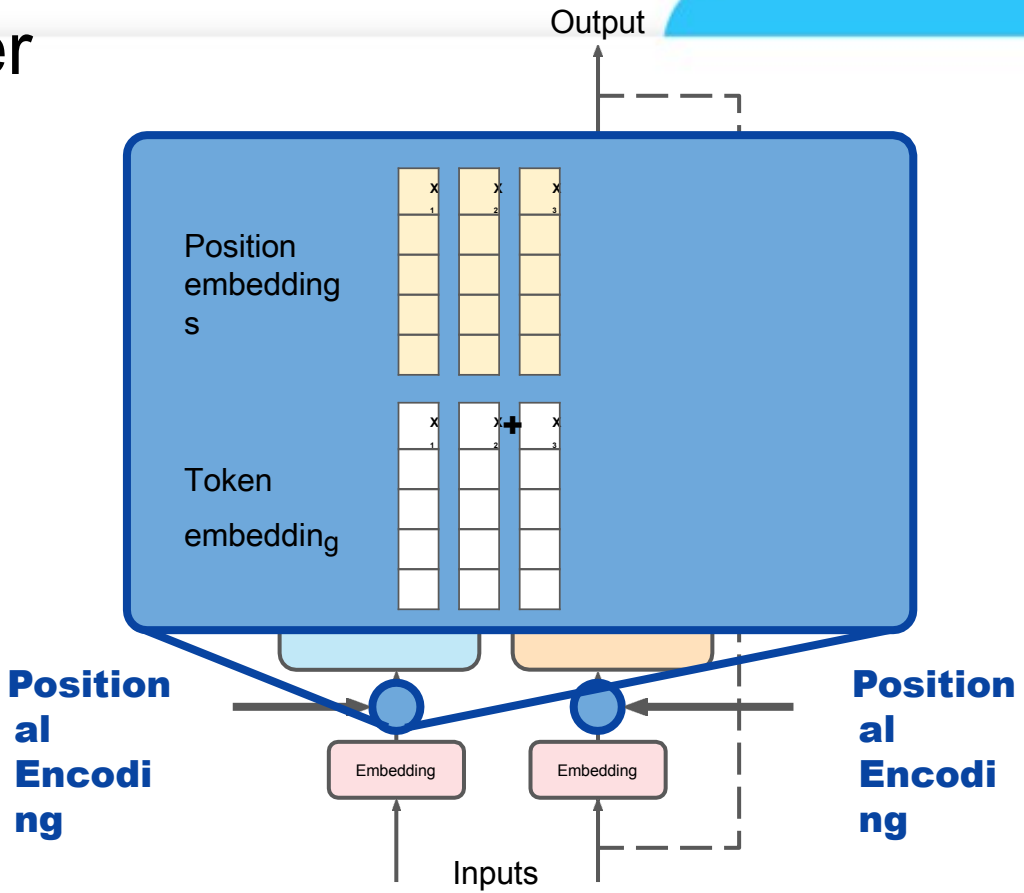
Transformers



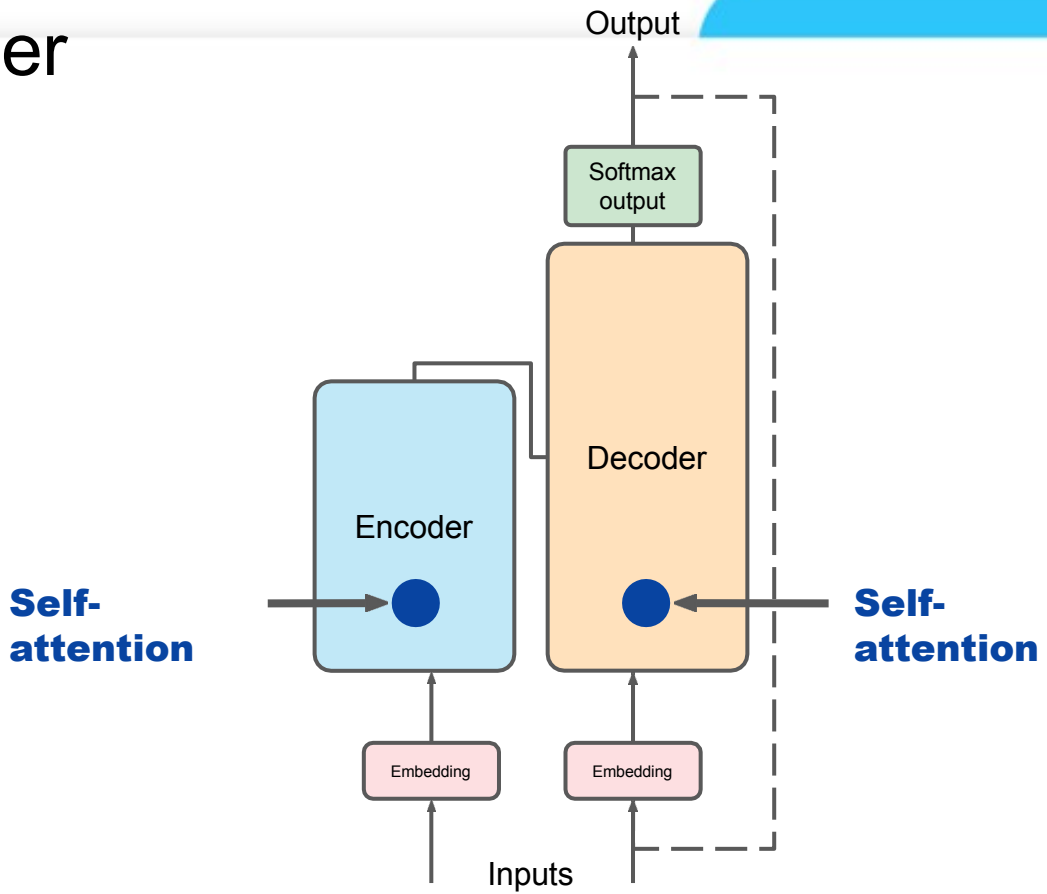
Transformers



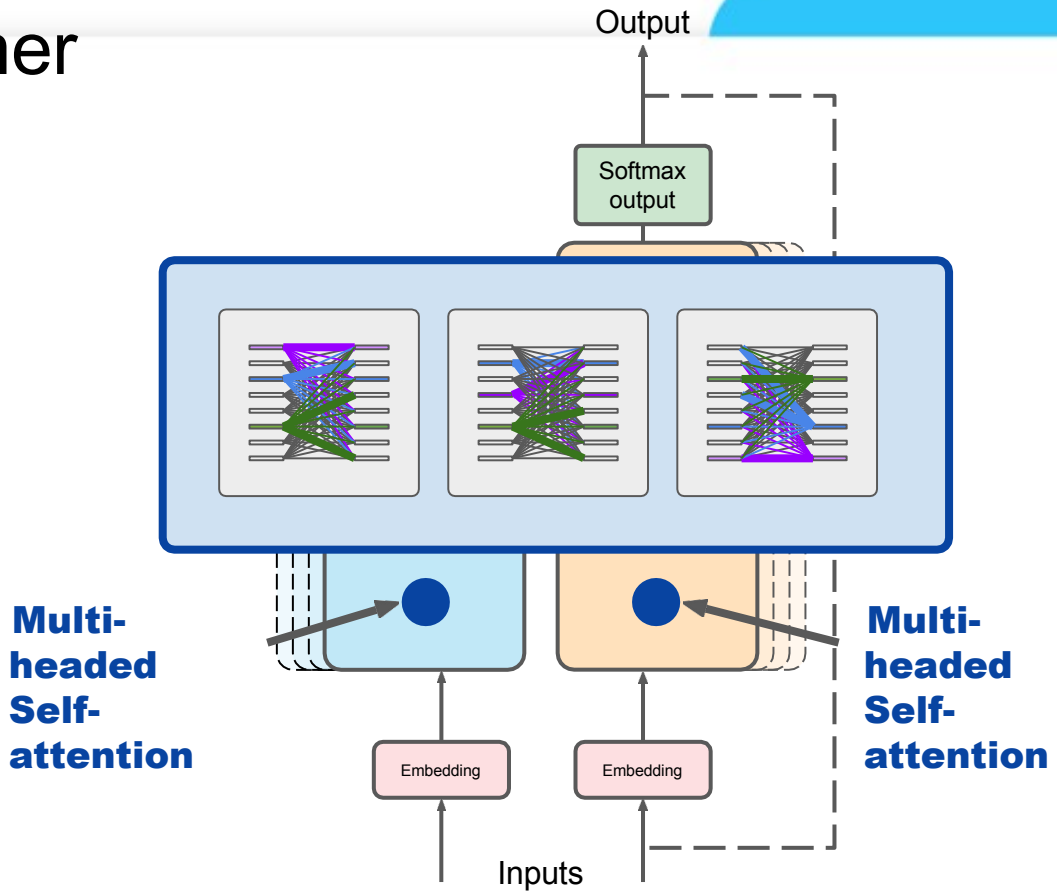
Transformers



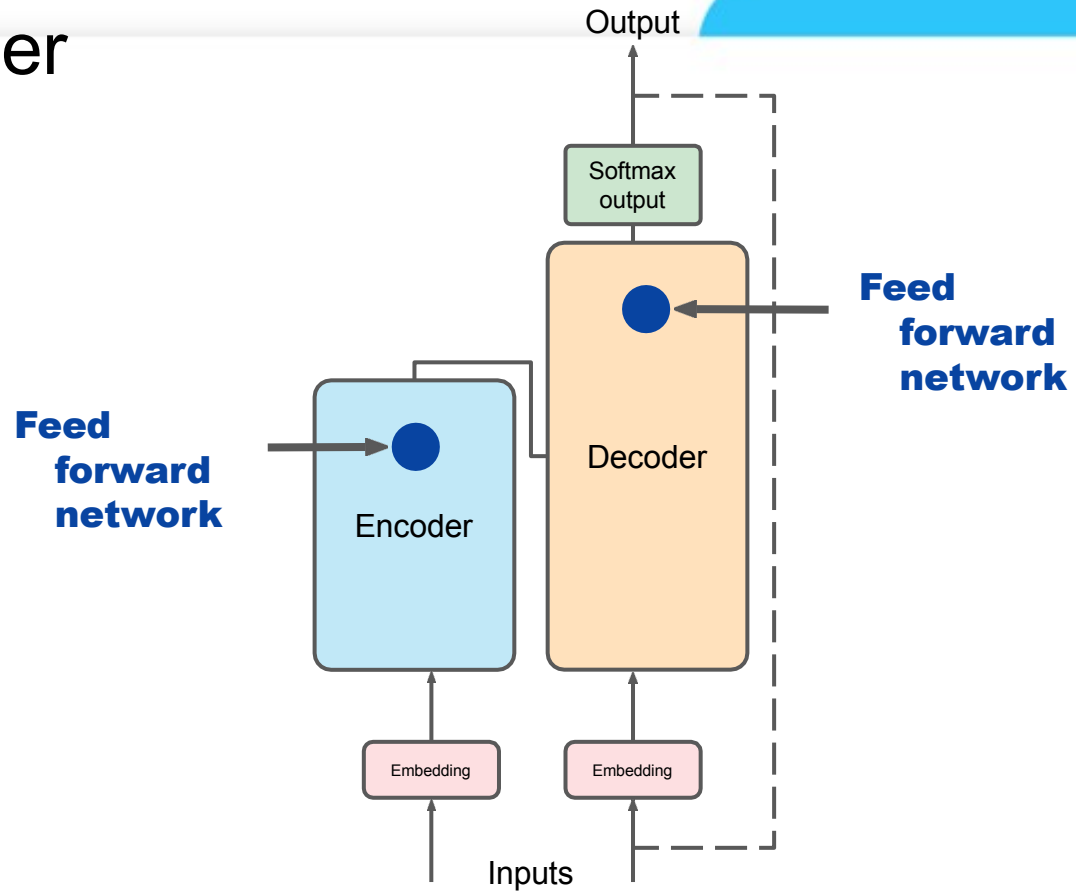
Transformers



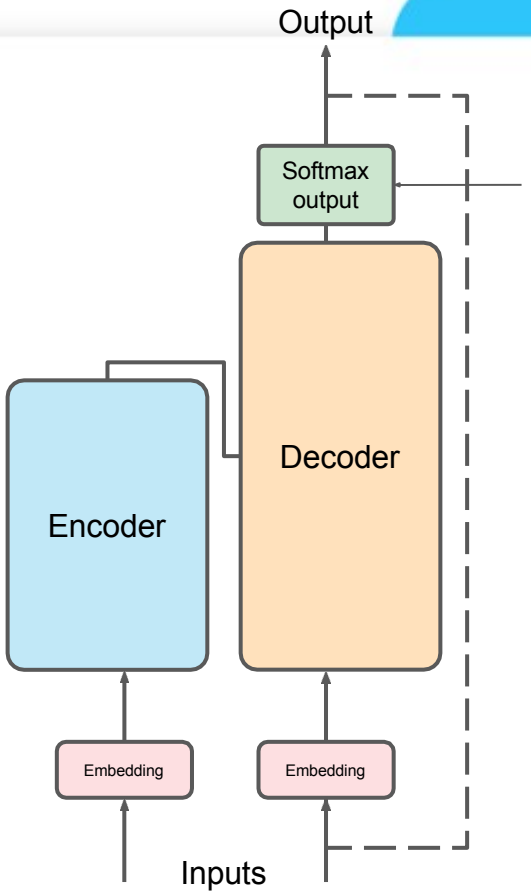
Transformers



Transformers

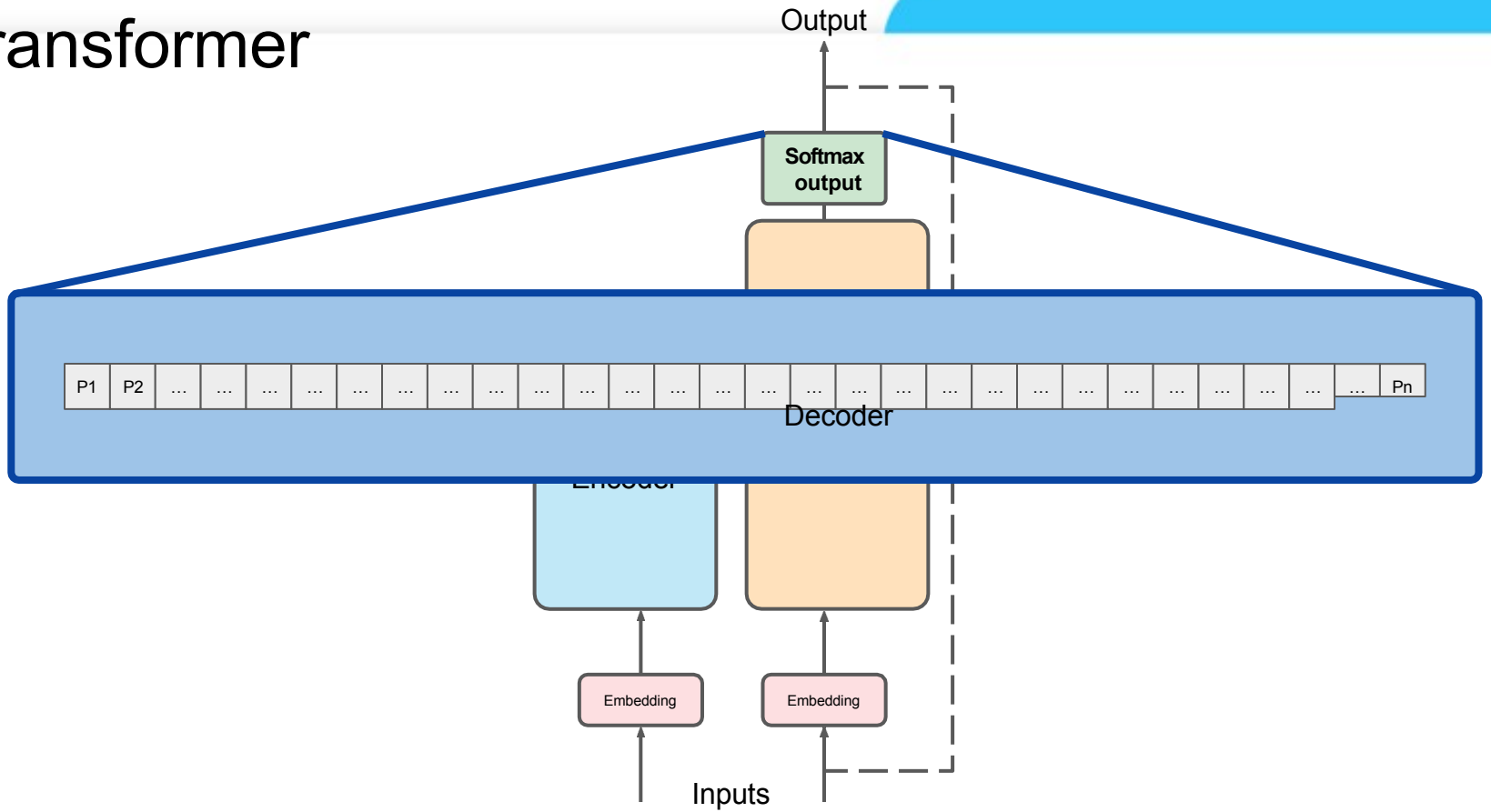


Transformers



Softmax output

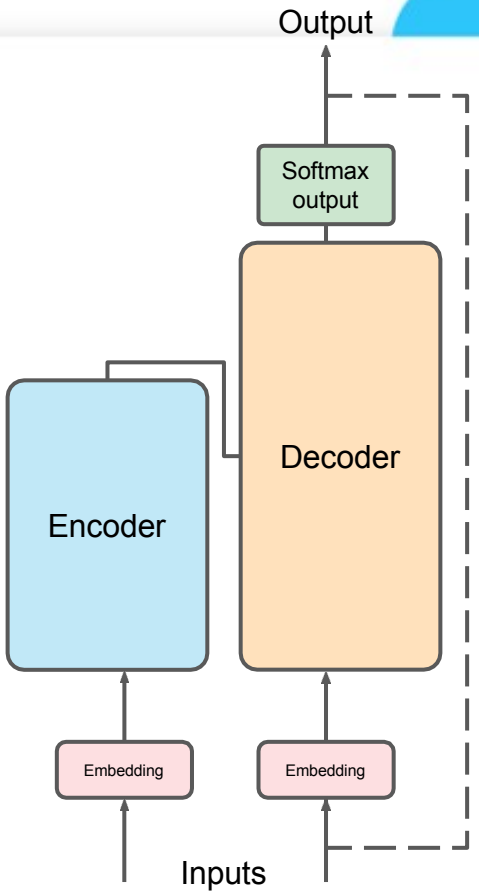
Transformers



Transformers

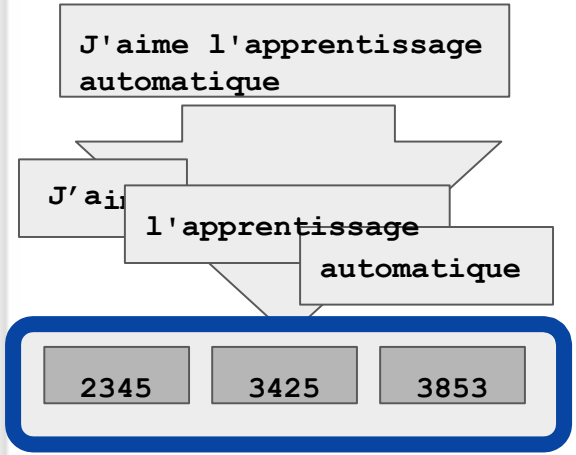
Translation: sequence-to-sequence task

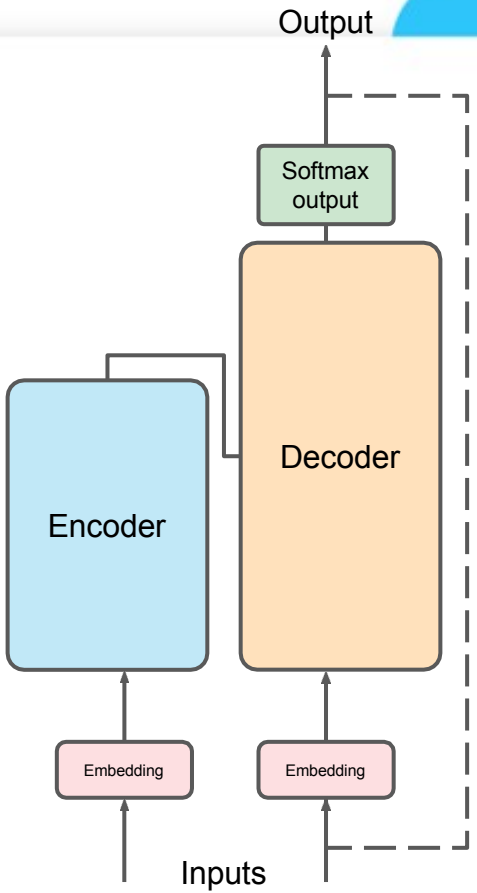
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J'aime l'apprentissage automatique
```



Transformers

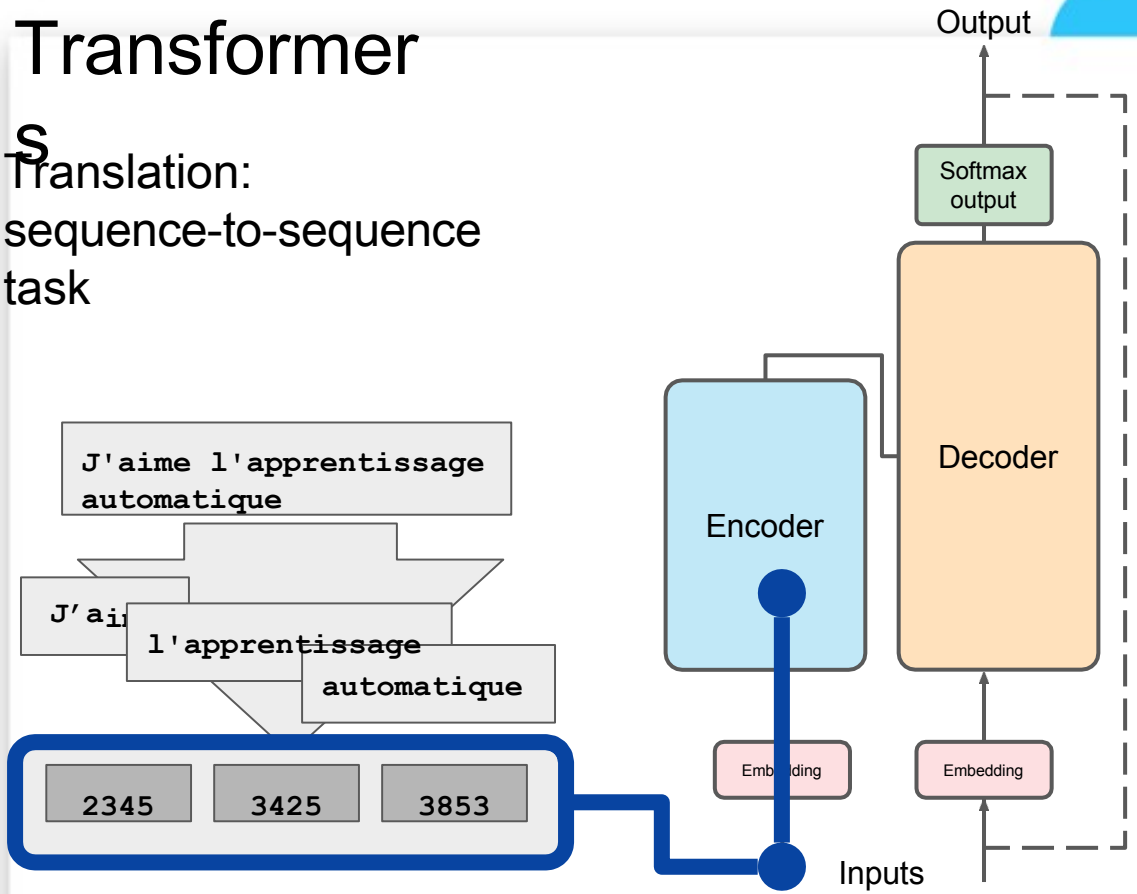
Translation: sequence-to-sequence task





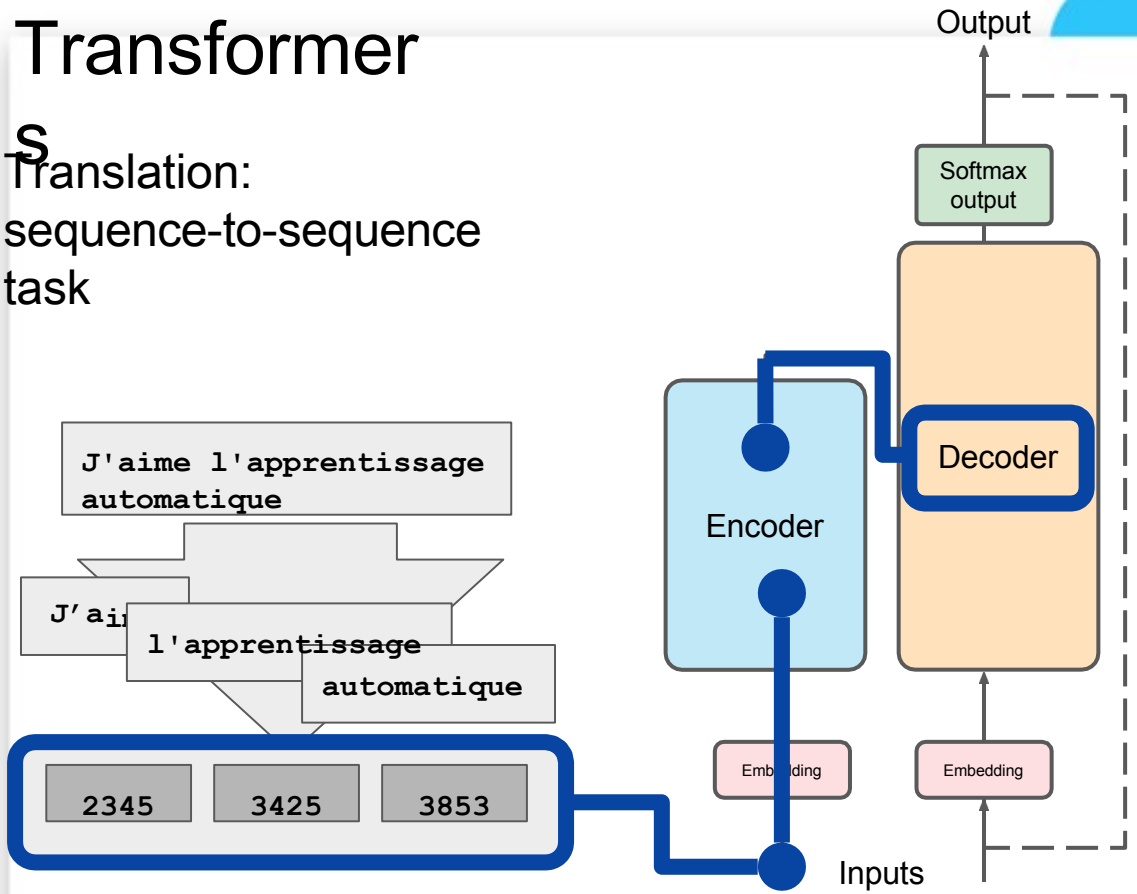
Transformers

Translation: sequence-to-sequence task



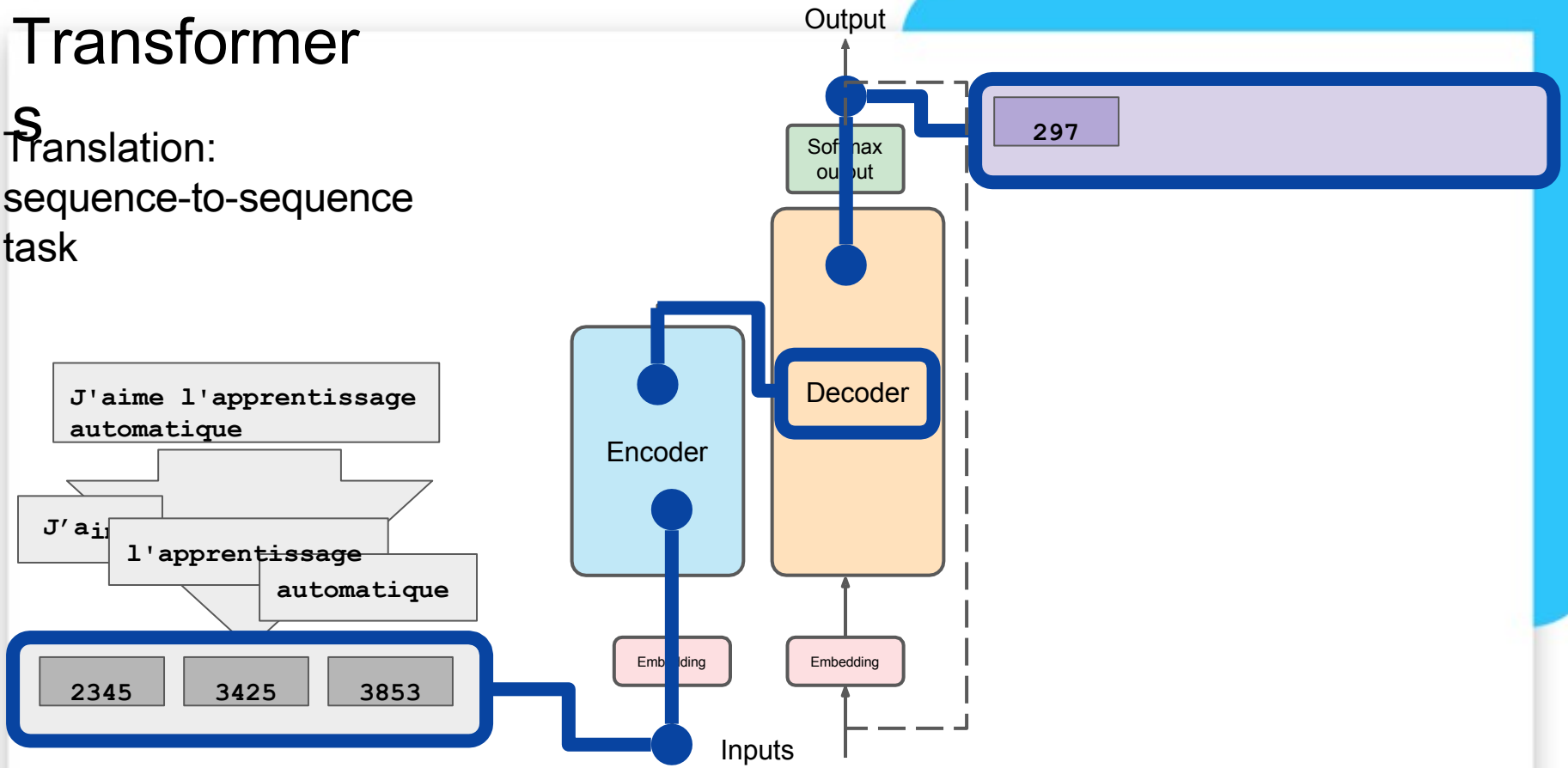
Transformers

Translation: sequence-to-sequence task



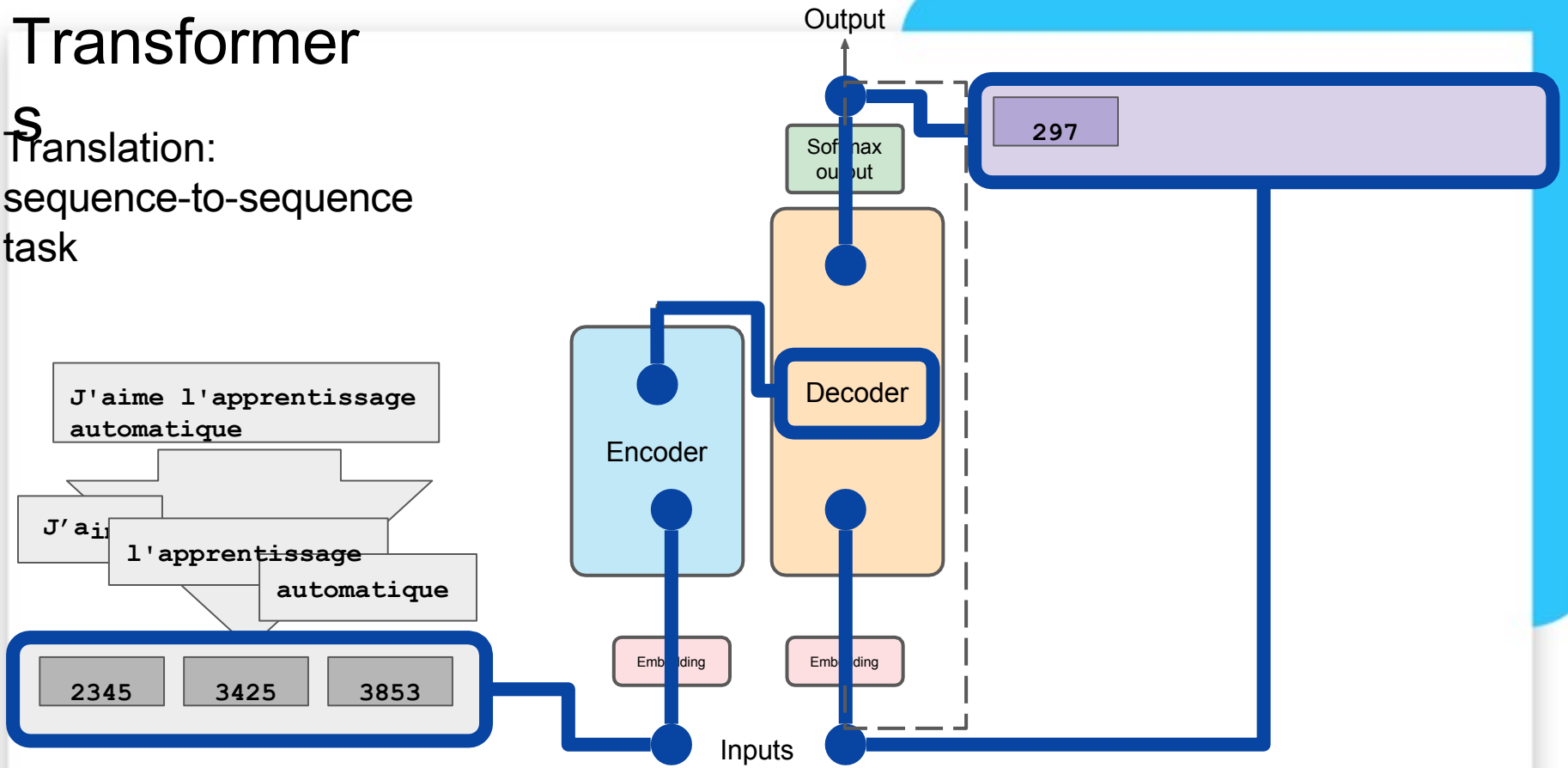
Transformers

Translation: sequence-to-sequence task



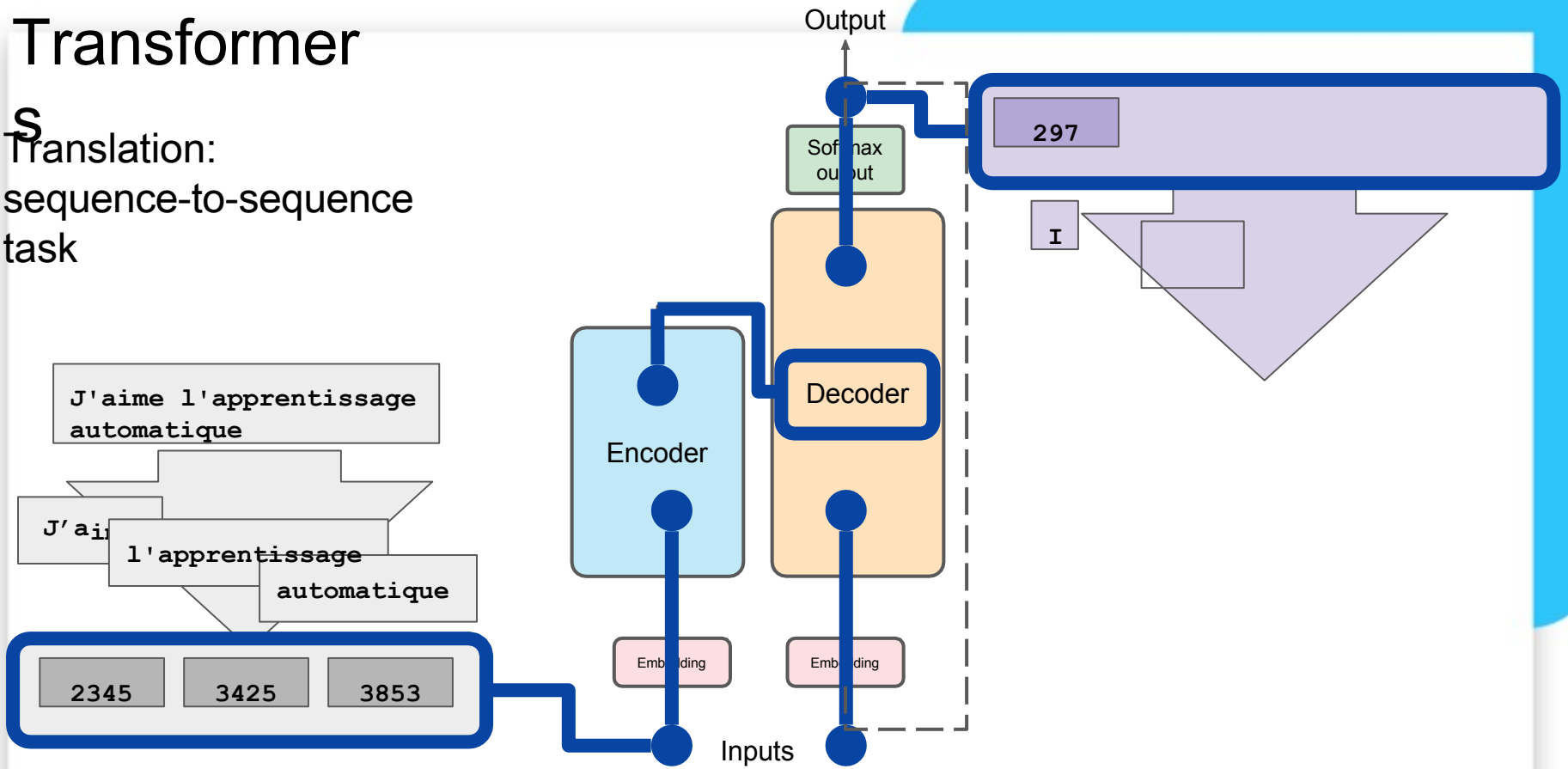
Transformers

Translation: sequence-to-sequence task



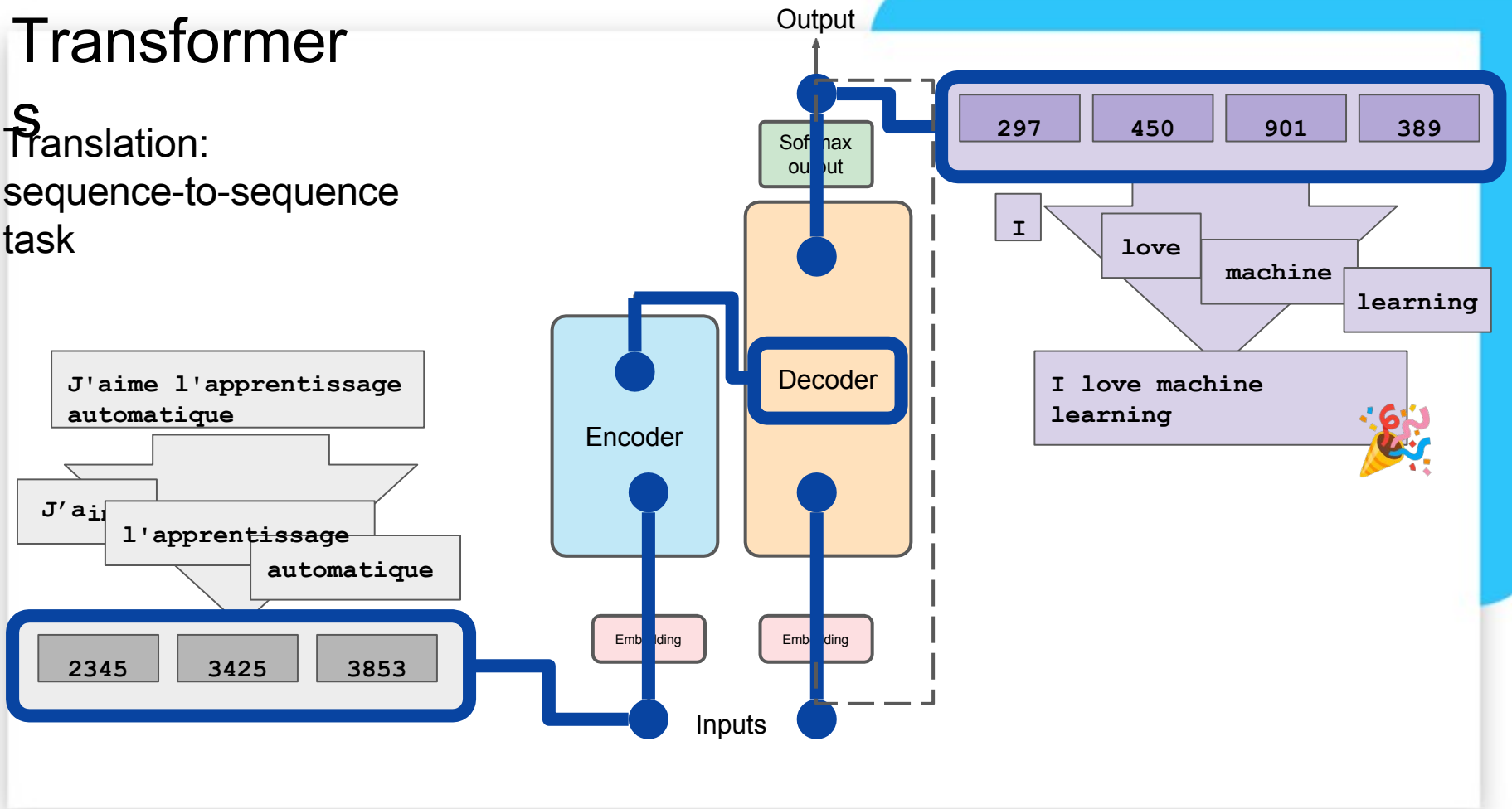
Transformers

Translation: sequence-to-sequence task



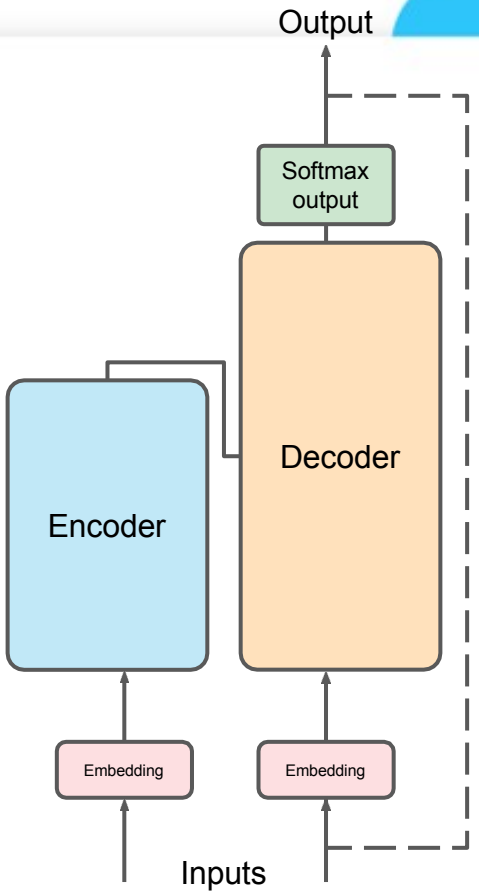
Transformers

Translation: sequence-to-sequence task



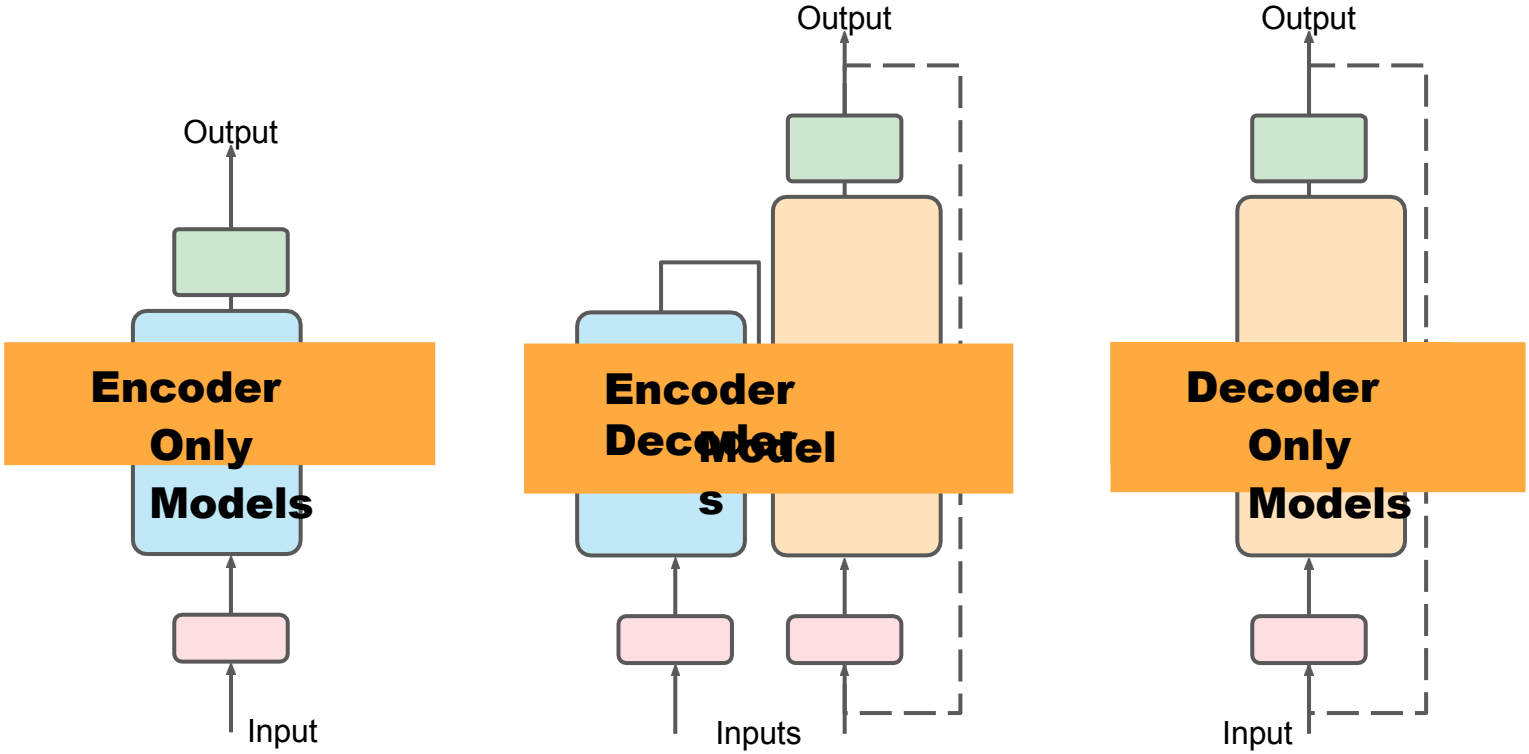
Transformers

Encoder
Encodes inputs (“prompts”) with contextual understanding and produces one vector per input token.



Decoder
Accepts input tokens and generates new tokens.

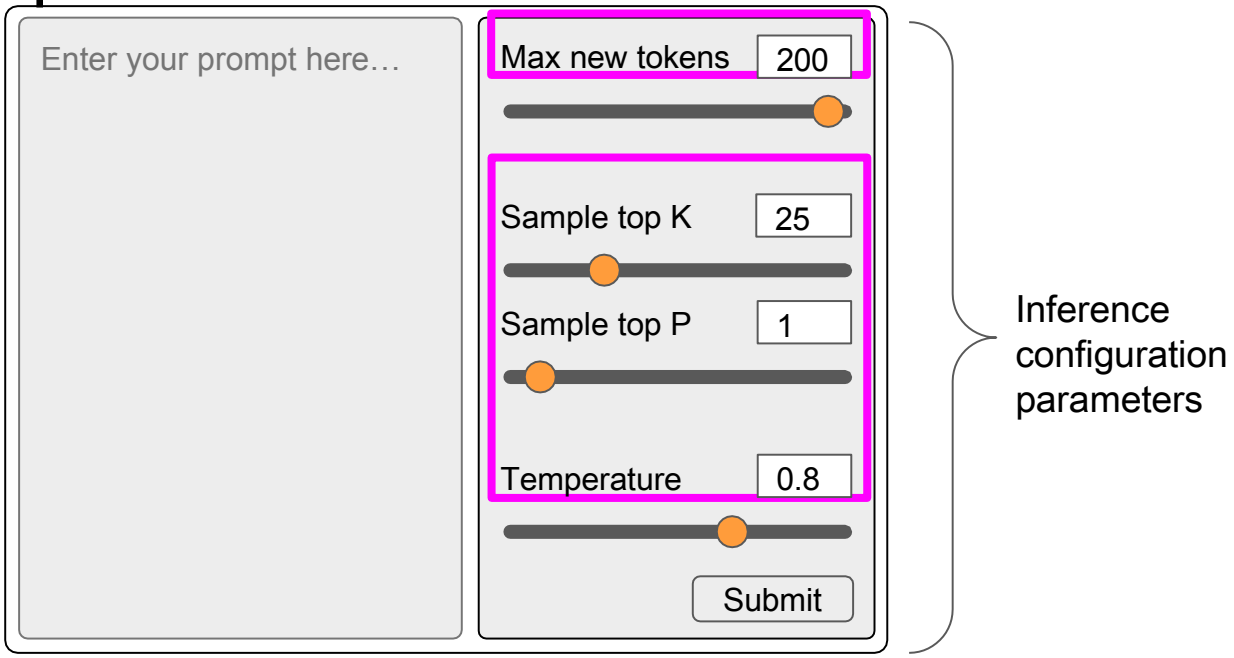
Transformers



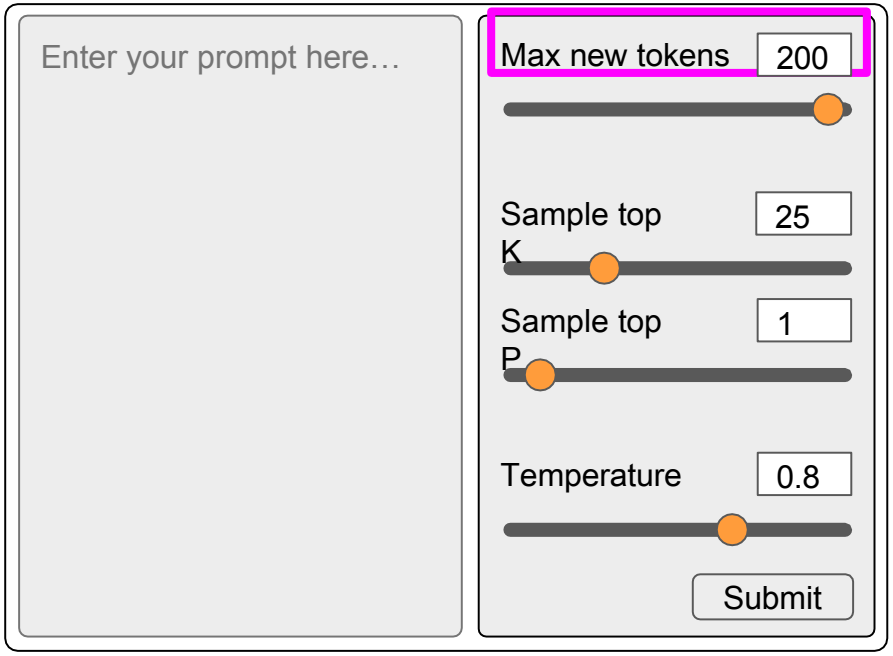
Jupyter

Generative configuration parameters for inference

Generative configuration - inference parameters



Generative configuration - max new tokens

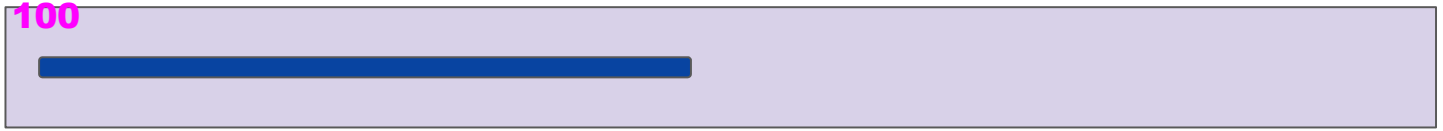


Max new tokens

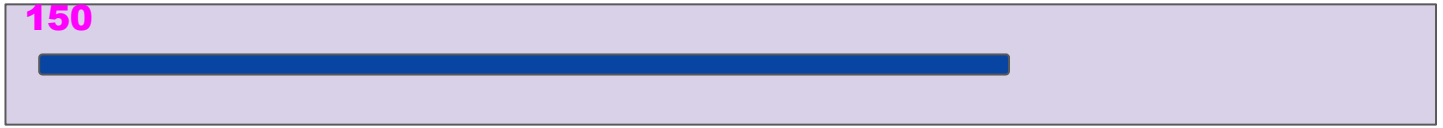
Generative config - max new tokens

Softmax output

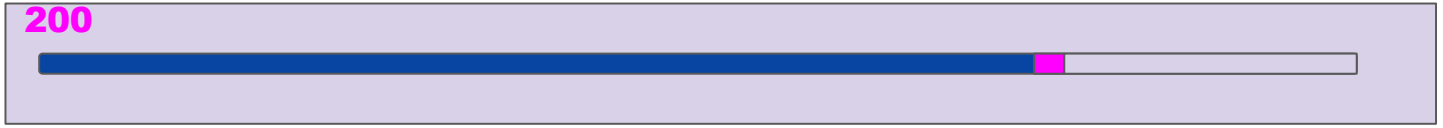
max_new_tokens = 100



max_new_tokens = 150



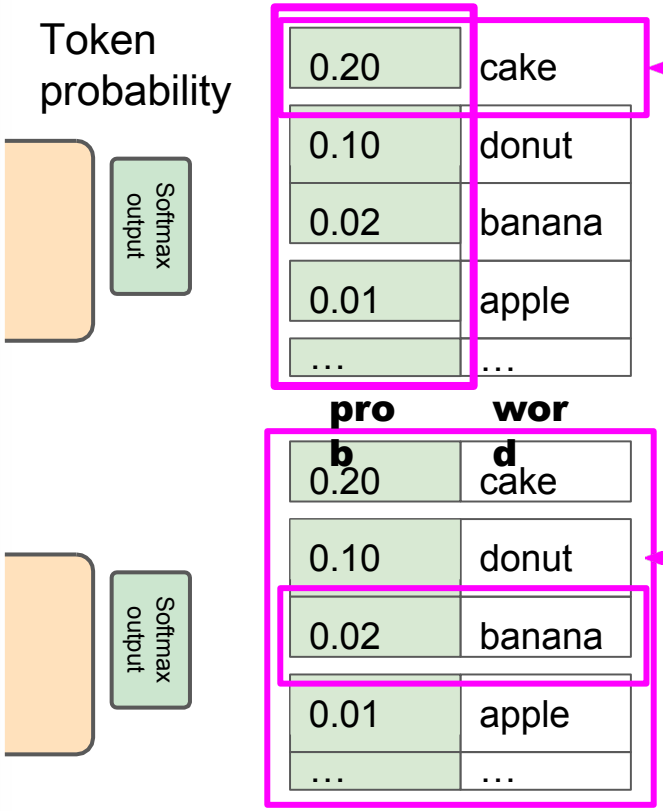
max_new_tokens = 200



Generative config - max new tokens



Generative config - greedy vs. random sampling

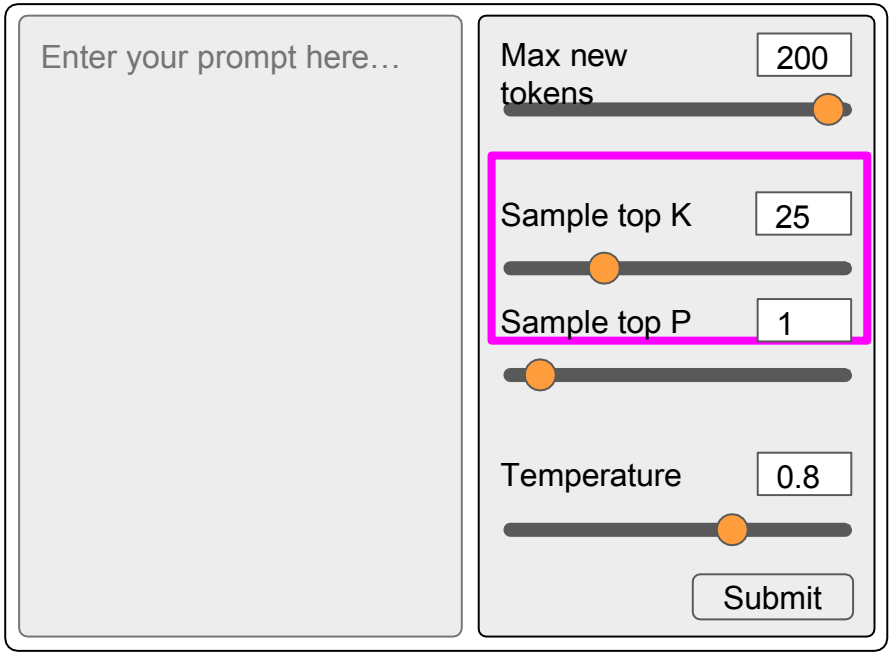


***greedy*: The word/token with the highest probability is selected.**

random(-weighted) sampling: select a token using a random-weighted strategy across the probabilities of all tokens.

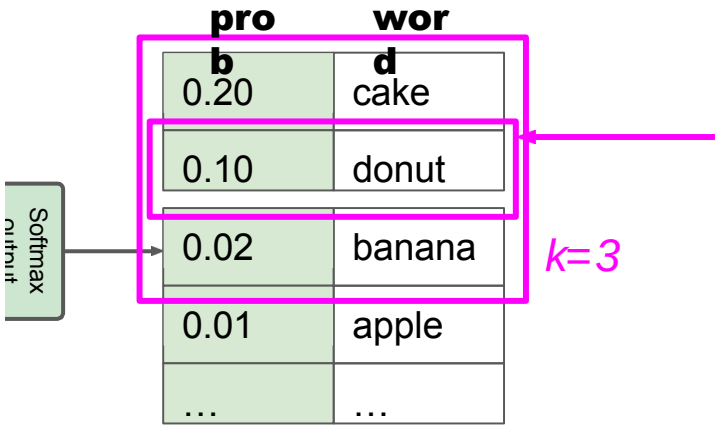
Here, there is a 20% chance that 'cake' will be selected, but 'banana' was actually selected.

Generative configuration - top-k and top-p



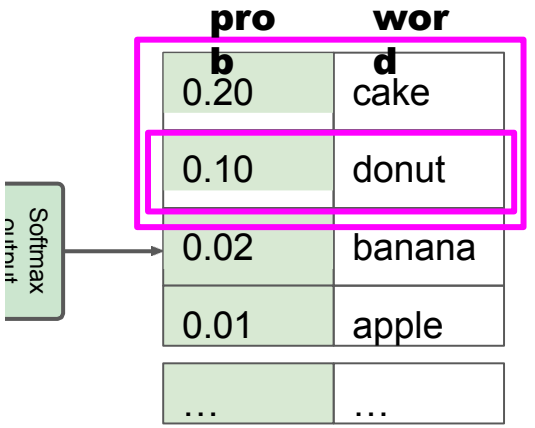
Top-k and top-p sampling

Generative config - top-k sampling



top-k: select an output from the top-k results after applying random-weighted strategy using the probabilities

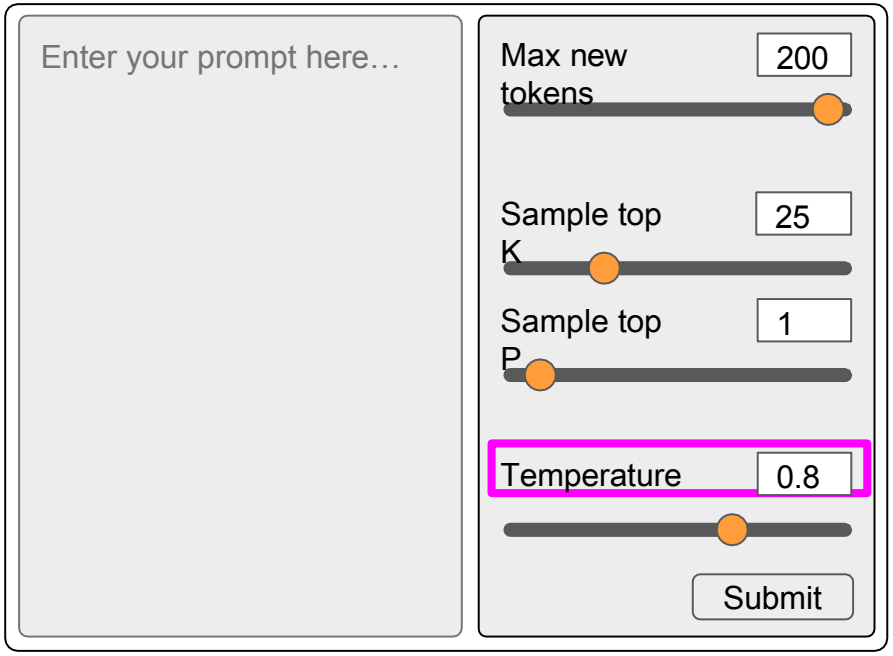
Generative config - top-p sampling



top-p: select an output using the random-weighted strategy with the top-ranked consecutive results by probability and with a cumulative probability <= *p*.

p = 0.30

Generative configuration - temperature



Temperatur
e

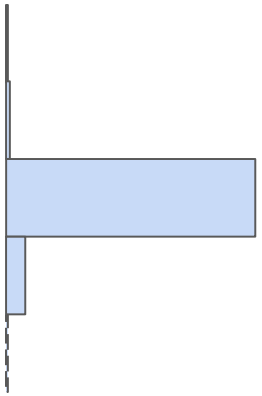
Generative config - temperature

Temperature setting

Softmax output

Cooler temperature (e.g <1)

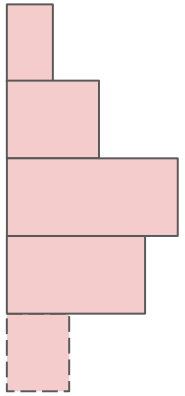
prob	word
0.001	apple
0.002	banana
0.400	cake
0.012	donut
…	…



Strongly peaked probability distribution

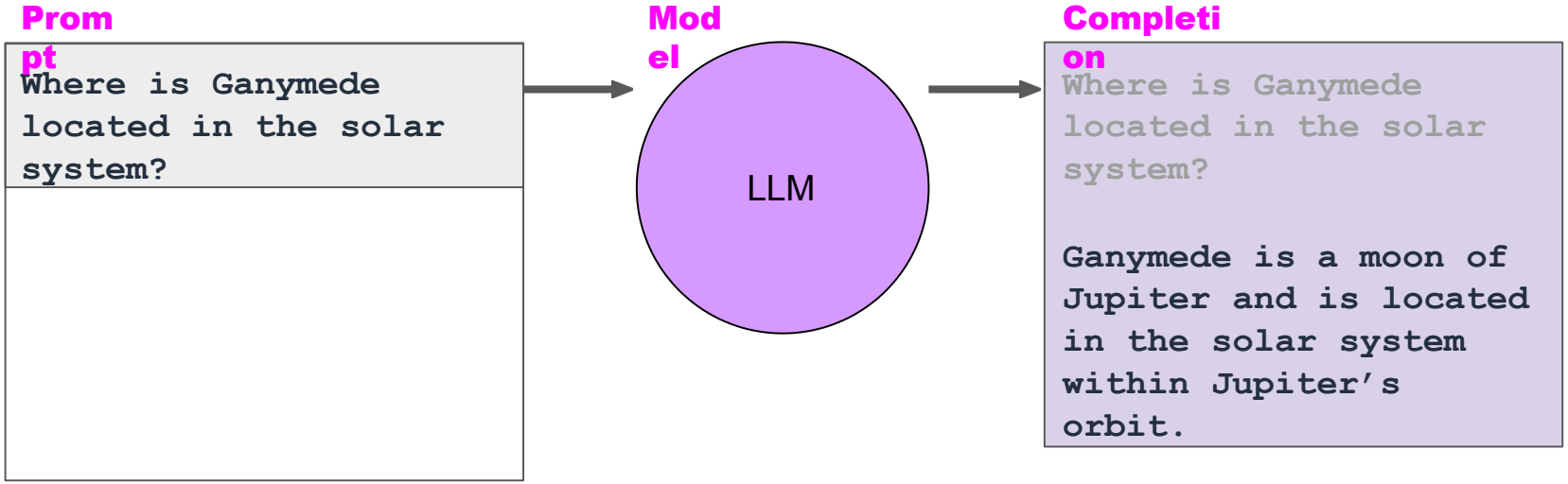
Higher temperature (>1)

prob	word
0.040	apple
0.080	banana
0.150	cake
0.120	donut
…	…



Broader, flatter probability distribution

Prompting and prompt engineering



Context window: typically a few thousand words

GenAI Concepts

Token

A token is the smallest unit of text that a language model recognizes, such as a word, part of a word, or even individual characters.

Prompt

A prompt is the input provided to a language model, consisting of text or instructions that guide the generation of a response.

Completion

Completion refers to the response generated by a language model in response to a prompt, completing the text according to the provided context.

Prompt Window Size

The size of the prompt window is the maximum number of tokens that a language model can process at once, including both the prompt and the completion.

Inference

Inference is the process by which a language model generates responses or predictions based on the input received, applying the knowledge acquired during training.